RESTORING UTILITY FUNCTION OR RISK PREFERENCE

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by

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Declaration

All sentences or passages quoted in this project dissertation from other people's work have been specifically acknowledged by clear cross referencing to author, work and page(s). I understand that failure to do this amounts to plagiarism and will be considered grounds for failure in this module and the degree examination as a whole.

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Abstract

The paper is mainly to solve the inverse portfolio problem based on two different risk theories which are Expected Utility Theory (EUT) and Mean Deviation Theory (MDT). In the two theories, investor preferences are described respectively in the form of utility function and general deviation. As it is non-trivial to establish exact utility function or general deviation in the practical market, the purpose of the work is to restoring expected utility function and deviation measure value over given time periods through modelling of the different theories. The result can be used to predict reasonable portfolio recommendation to each individual investor based on the history portfolio he or she holds.

All the prediction will be based on the history trading data in the market. Rather than the stock's closing price data on each time period, the interest rate of each period is the focus for prediction, because it denotes the change of the stock price on each time period. Therefore, the work's analysis based on the interest rates over chosen time periods.

In the paper, the risk aversion case is only considered. In the risk aversion case, for the inverse problem, the expected restoring function would be strictly increasing concave function. The paper work is trying to find the restoring function value over chosen time periods using the portfolio value the investor holds on the basis of the portfolio weights choice set of chosen number of assets.

Based on the randomness of stock prices in the practical market, and risk aversion assumption, we propose the definition of stochastic dominance (SD) to relate it to the expected concave function and Schur-convex function respectively. Both stochastic dominance and corresponding function can represent a reasonable decision between two choices of the portfolio weights. In expected utility theory, the restoring function should be strictly increasing concave function, however, in mean deviation theory; the restoring function should be Schur-convex function. Both restoring function can represent the preference between two choices of the portfolio weights. It can be noticed that the property discrepancy of functions restored in two theories is consist with their own optimization statements respectively.

Using the EUT and MDT, it is possible to get several numbers of restored functions over chosen time periods respectively. Traditionally, we would prefer to choose the minimum of the restored expected utility functions and the maximum of the restored general deviation measures functions. The reason for this choice is different optimization properties in the distinct two theories.

Traditionally, the optimization is to maximize expected utility function with respect to portfolio weights in EUT and minimize general deviation measure regarding to portfolio weights in MDT.

The main aim of the introduction of concept of stochastic dominance is to make sure the existence of the solution of the inverse portfolio problem using EUT and MDT respectively. For the purpose of better accuracy of the representation for reasonable behaviour, second-order stochastic dominance (SSD) is introduced, which would be used to guarantee the sufficiency of the existence of inverse problem using mean deviation theory and can be generalized to expected utility theory.

The large scale of time series observations would follow the normal distribution according to the central limit theorem (CLT). Besides, the produced solution of the inverse portfolio problem would be function values over chosen time periods on portfolio values the investor holds, which can be regarded as a random variable. The random variable admits a set of random values over chosen time periods. Due to the concerns of the inverse portfolio problem is a choice of the portfolio weights of given number of assets, the time should be inconsequential in the set and the random values in the set can be sorted traditionally according to ascending order. In the modelling, the time independence of the chosen portfolio values should be the basic hypothesis.

To hold a proper optimal choice of the portfolio weights, the model of popular Markowitz portfolio selection problem is introduced and implemented before the modelling of the inverse portfolio problem. Based on the proposed basic assumption and given optimal portfolio values, two modelling based on EUT and MDT are presented and carried out separately.

It should be noted that two implement method of modelling using MDT is presented. In the first method, the derivatives of the general deviation measure are produced over each time period, which is the function of a real variable denoting time periods. And then it is simple to calculate the definite integral within an interval from 0 to 1 in the real line using the definition of mixed CVaR-deviation. It can be noticed that formulating the derivatives of the general deviation measure and graphing its functions with respect to the time periods is convenient. However, the first implement way has shortcomings. It may produce several numbers of deviation measures; hence, the second modelling approach is introduced using MDT. According to the known Proposition […], the method of implement will definitely generate unique general deviation measure function and the formulation and implement of the better modelling approach is presented in the recommendation for improvement part in Mean_Deviation_Model.

In the thesis, the time series independent identical distribution (IID) needs to be tested before model test. The carrying out of the modelling of the inverse portfolio problem is expected concave utility function using EUT and Schur-convex deviation measure function using MDT. It should be noted that the relationship between SD and the derivative of the Schur-convex deviation measure is the same as the connection between SD and concave function according to a simple criterion in the theory of Schur-convex function. Hence, the expected graph of the derivatives of the general deviation measures should be concave function. However, in the case that exists several deviation measure functions, the maximum value is the concerns in the MDT. Therefore, if the produced derivative graph looks like an upper semicircle within the interval between 0 and 1, it should be consistent with the theoretical prediction of MDT.

In the conclusion section, it consists of two parts. Based on the result of the modelling of the inverse portfolio problem using EUT and MDT respectively, the first part is to predict future trading strategy for the prior individual investor so that he or she can enjoy more opportunity to invest rationally. It is sure that there exist weak points in the modelling presented, and then the improved method is also proposed.

Key Words: inverse portfolio problem, data concern, assumption, SD, SSD, EUT, MDT,

Chapter 1
Problem Proposing

In this chapter, the need for investment portfolio and rational investment behaviour is presented; following which, the mathematical formulas and necessary conditions is introduced for the definition of investment portfolio. In the risk tolerance existing context, merely risk-aversion case is considered throughout the whole thesis, and then the inverse portfolio problem is proposed formally.

1.1 Investment portfolio and rational investment behaviour

Investment portfolio is an important financial term and refers to any collection of financial assets such as stocks, bonds and cash, in the thesis, common stocks in the market is dealt with. It is a very useful instrument in the risk tolerance existing context because it is originally appeared as a risk-adjusting tool. It can produce the guide for the rational behaviour of individual investors. Traditionally, when determining a proper choice of the portfolio weights based on chosen numbers of portfolio weights, the investor aims at maximizing the expected return and minimizing the risk, which is maximizing the expected utility function with respect to portfolio value in EUT and minimize the general deviation measure regarding to portfolio value in MDT.

In the realistic financial trading market, it is convenient to access several number of assets, say, N assets, and the history stock closing prices over chosen number of time periods, say, T time periods. Due to the focus on the change of stock prices in the trading market, the rate of return of history stocks should be processed instead of their trading prices for the purpose of the trading prediction. The formulation of the portfolio return is based on the rate of returns values on chosen assets over T time periods.

The mathematical portfolio formulation is presented as follows:

- Assets: 0 , 1, 2, … , N
- Rate of returns: $r_0, r_1, \ldots, r_N$
- The portfolio return: $X = \sum_{i=0}^{n-1} \lambda_i r_i$

The portfolio return is one-to-one correspondent to choice of the portfolio weights vector $[\lambda_1, \ldots, \lambda_n]$.

1.2 Introduction to Risk-aversion

As the investment portfolio is a way to attempt to reduce the uncertainty while exposed to the real risky market and the interested context in the paper should be risk tolerance existing circumstances. As investment portfolio is studied, that being not insensitive to risk is the concerns in the paper. It means that individual investors would care whether they took the guaranteed payoff or the gamble. Then they may have different risk attitudes, risk-avoiding or risk-loving in the financial trading market. In the thesis, the risk-aversion case is only considered.

Risk-averse is a concept based on the behaviour of humans in the risk-tolerance existing context. It refers to the reluctance of a person to accept a bargain with an uncertain payoff rather than another bargain with more certain, but possibly lower, expected payoff. Clear explanation drawing can be seen in the following example. Consider two scenarios trades; one is a bet, the other is a certainty trade which would get guaranteed payoff. If an investor is insensitive to risk, the expected payoff produced from the left part and right part is equal to the same value, say, $50.

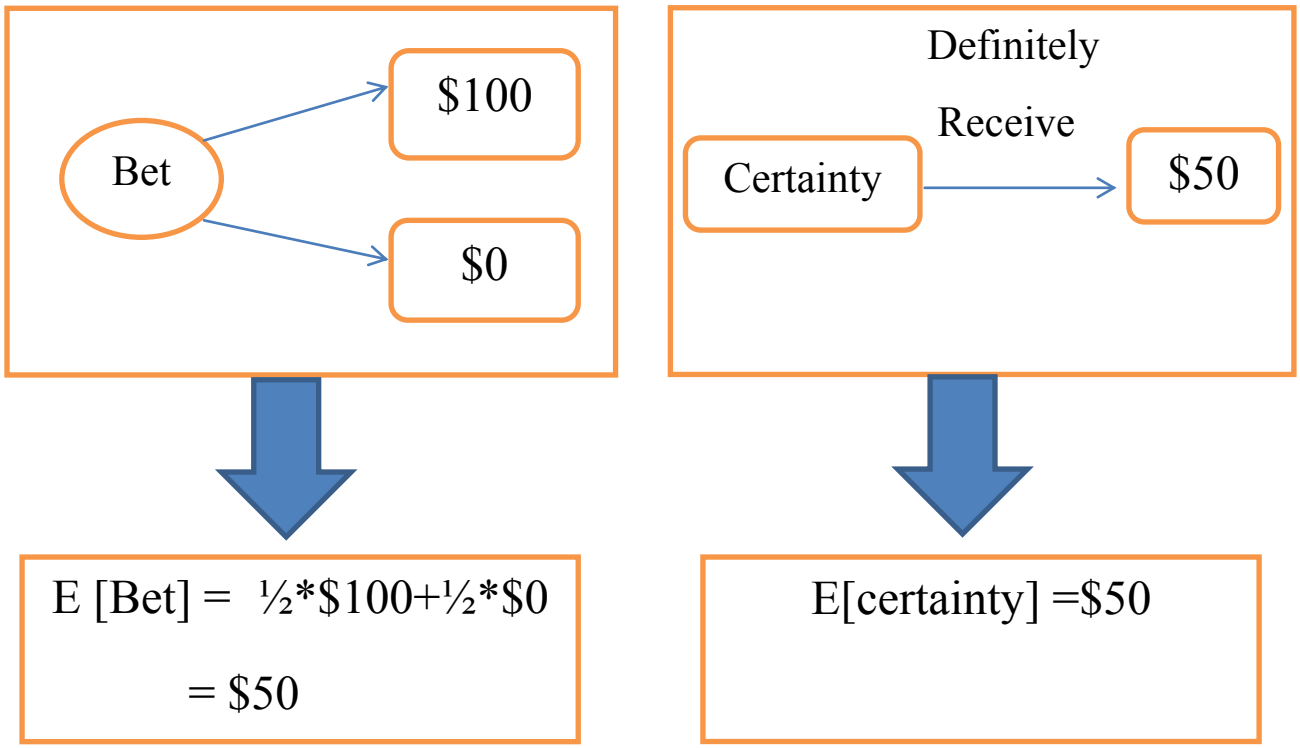


Figure_1

However, the interested context in the thesis is the risk-aversion case, in the above example, instead of taking the same value, in the right part figure; the investor would accept a certain payment of less than $50(say, $40), rather than taking the gamble in the left part figure even if the return is greater than the certain payoff $40. Therefore, the graph of function denoting the risk preference should be concave function in the risk-averse circumstance, the theoretical graph looks like the figure_2.

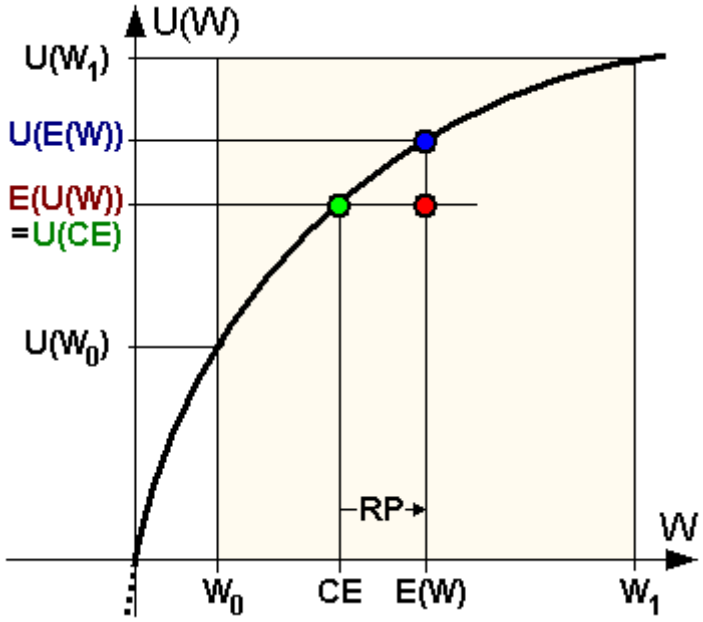
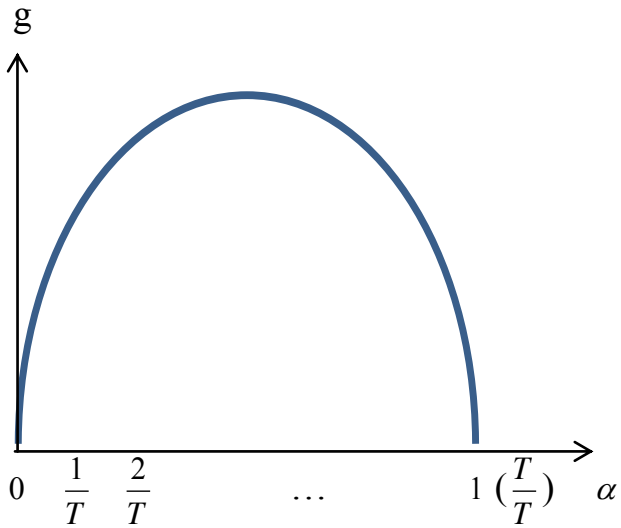


Figure_2

In the figure_2, E (W) (expected value) denotes the expected payoff in the gamble scenario(uncertainty); CE(certainty equivalent) represents the return that the individual would accept instead of the bet; RP(risk premium) is the difference between E(W) and CE.

It can be noticed that for the risk-averse individual, RP becomes positive. From the risk preference function In Figure_2, the investor prefers to choose the green point which admits the value less than expected payoff in the gamble. As the restored value of CE equal to the expected utility function with respect to choice of the portfolio weights, the solution of the inverse portfolio problem using EUT should be concave function while the result of the inverse problem in the MDT is Schur-convex function which can be deduced in the section of 5.2.

It is easily to graph the derivative function of the general deviation measure in MDT with respect to the random variable denoting time periods. The theoretical drawing of the derivative function of time periods using MDT should be a semicircle, the figure is as follows.



1.3 Proposal of inverse portfolio problem

In the problem related to investment portfolio, the study object is the portfolio weights. In the modern portfolio theory, the portfolio weights can be easily obtained using the popular Markowitz portfolio selection problem. However, the predicted portfolio weights is only a theoretical concept which is hardly to be as a proper guid of the reasonable investor behaviour. [...] It is necessary to introduce better theory which is more closer to Market portfolio estimates, such as the expected utility theory(EUT) and the Mean_Deviation measure theory (MDT). The common advantage of the two theories is that restoring risk preference function based a bench of financial instrument(stocks, assets, .., etc) is possible and convenient to implement because the assets type and stock prices can be exposed to almost all individual investors.

The inverse portfolio problem is the opposite process of portfolio optimization problem. The aim of the inverse problem is to restore the risk preference function which takes the form of the expected utility function using EUT and the general deviation measure using MDT respectively.

Both restored risk preference functions' arguments in the EUT and MDT are an optimal choice of the portfolio weights based on chosen number of portfolio weights, say, N time periods. which is already held by an investor. The given

choice of the portfolio admits a set of random portfolio values over chosen number of time periods, say, T time periods. Besides, to guarantee the continuity of the restored risk preference function, all choices of the portfolio weights based on N numbers of assets should form a bounded and convex set. That is, the line segment connecting any two elements of the set should be within the set.

Starting from the above known conditions, in the EUT, the objective of the inverse portfolio problem is to find a utility function $u(X) \in U$ such that (s.t.) the given choice of portfolio weights X^* is optimal for $\max_{X \in V} E[u(X)] = E[u(X^*)]$ with u , for U is a set of strictly increasing and concave function. By analogy, in the MDT, the aim of the inverse problem is to restore a deviation measure $D(X) \in W$ s.t. X^* optimal for $\min_{X \in V} D(X) = D(X^*)$ with D, for D is a set of Schur_convex functions.

Chapter 2
Problem Analysis

In the chapter 2, the section can be divided into five main parts, firstly, analyse how to produce an appropriate value as the given condition for inverse portfolio problem. Following that, data collection strategy is presented in the second part. Thirdly, the clear explanation for expected restored function using EUT and MDT is stated separately. Fourthly, the necessity of introducing stochastic dominance (SD) and second-order stochastic dominance (SSD) is presented. In the fifth part, how to use the numerical result of the inverse portfolio problem to provide the individual with a future trading strategy recommendation is presented.

2.1 Generate given optimal choice of portfolio weights

The inverse portfolio problem begins with the given optimal choice of portfolio weights and the optimal choice value should be specified properly so that the modelling of the inverse portfolio problem is implemented effectively. Markowitz Mean-Variance theory is modelled to generate an choice of portfolio weights which is the starting point of the work of the thesis. The Mean-Variance model can be classified as a quadratic programming problem which can be conveniently carried out using MATLAB.

2.2 Data objects collection

In the inverse portfolio problem, data collecting should be chosen properly. The interesting study object is the portfolio interest rate which can convey movement information of the stock price in the stock exchange. The history stock prices are available to every individual investor and the stock weekly closing price is the concerns in the thesis. Based on these price, the formula $R_t = \frac{P_{t+1} - P_t}{P_t}$ is used to produce the focus data of the thesis, which is the return of stock price, and then the chosen return of portfolio value can be obtained using formula $X = \sum_{i=0}^{n-1} \lambda_i R_i$. Besides, the dimensions of assets and time period should be chosen carefully. Traditionally, the chosen number of assets is less than the chosen number of time periods, say, N< T. The proposition 9 in the paper […] states that the maximum number of nonnegative portfolio weights would be N+1, and the value of choice of portfolio weights can be obtained using the same modelling theory only a little change. Therefore, N numbers of asset, T numbers of time periods, N=T-1 is the normal data dimensions chosen condition. Specifically, (N=3, T=4) is chosen in the modelling of EUT and (N=10, T=11) in the MDT.

2.3 Explanation of expected restored function in EUT and MDT

In the EUT, the risk preference function takes the form of general deviation measure while it is expected utility function using MDT. The restored function of the inverse portfolio problem should strictly concave function and Schur-convex function respectively. The explanation of different function property can be seen in (1.2) and (5.2). In the section, the argument of the restored function is highlighted while the detailed explanation of the restored function itself is presented in section (5.2).

It is clearly that stock prices over chosen time periods are random numbers in the realistic market. The solution of the inverse portfolio problem is the restored function and the argument of the function is portfolio values over chosen time periods. Being in accordance with practical trading market, the portfolio values should be of randomness and each choice of portfolio weights would correspond to a random value in single-time period. In the generalization of several chosen number of time periods, it will correspond to a set of random values. That is, the set of random values can be seen as values of a random variable over different time periods. From the perspective of the restored function that is the solution of the inverse portfolio problem, it is convenient to

see that the order of random values in the set is inconsequential because the key point in the inverse portfolio problem is only to use one choice of the portfolio weights, which denotes merely a random variable. Therefore, the argument of the restored function should be random variables. Distinct choices of the portfolio weights would represent different random variables.

It should be noted that there exists a very important distinguish point throughout the whole thesis. Due to all choice of the portfolio weights based on the same set of assets, the formed random variables should be related to each other. However, it is totally the different case from the perspective of solving process of the inverse portfolio problem. In the inverse portfolio problem, the study object is simply one random variable and the interest data is the set of random values over chosen time periods. The aim of the inverse portfolio problem is to find the value of the restored function on each random value.

Assume the restored function is denoted by f(X), and X is choices of the portfolio weights and represent different random variables. Each choice corresponds to a given value of portfolio vector $\lambda = (\lambda_1, \ldots, \lambda_n)$, that is $f(X) = f(\lambda_1, \ldots, \lambda_N)$. However, $f(X) \equiv f(X_1, \ldots, X_T)$. Clear illustration of the augment of the restored function and discrepancy of the knowledge mentioned above is as follows. In the figure, assuming two random variables X* and X**.

(1) For each specified argument of the restored function, say, X* or X**.

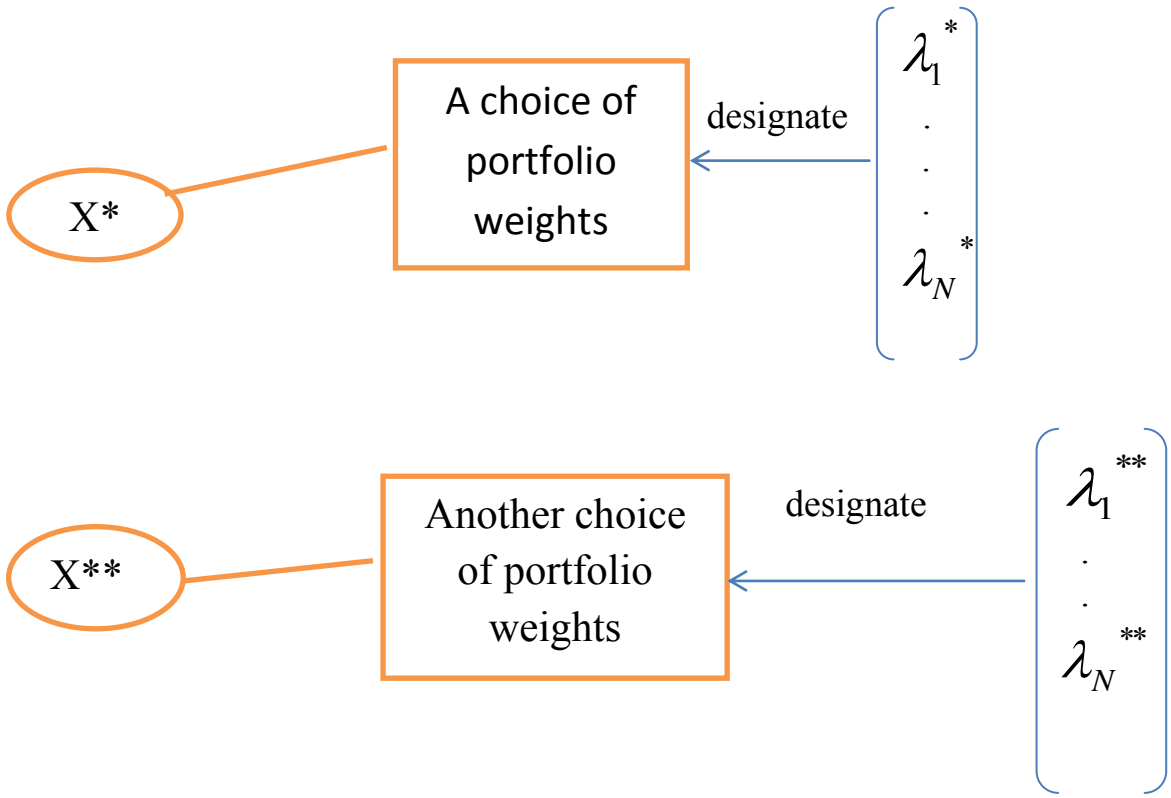


Figure_3

(2) In the inverse portfolio problem, one specified argument, i.e., an optimal choice of the portfolio weights is the concerns in the solving process of the problem, say, X* or X**.

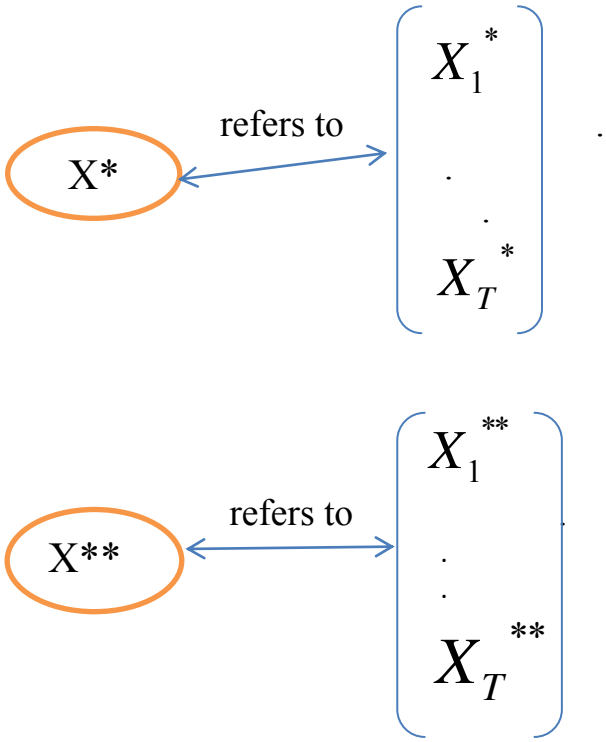


Figure_4

(3) When modelling the inverse portfolio problem, it is expected to restoring the risk preference function f(X) over T time periods, that is, the restored function is f ($X_1, X_2, \ldots, X_T$). Using EUT and MDT, the argument of the restored function is the same, i.e., X. However, only one value of X is the interesting object for restoring process, say, X*. It means, in the solving process of the inverse portfolio problem, no matter in EUT or MDT, the input argument only one value on each time period, say, time period t = 1, we try to use the input argument X_1^* to restore risk preference function ‘ f ’ such that f (X_1^*) is the maximum or minimum value of f (X_1) . Similarly, it can be generalized to any other time period t. Traditionally, in the modelling, the restored function over chosen number of time periods is the piecewise function.

It should be noted that the given known condition for the inverse portfolio problem should be optimal choice of the portfolio weights, which can be

obtained from the Markowitz portfolio selection problem based on Mean-Variance Theory. Say, X*, then, the argument of the restored function is $X^* \equiv (X_1^*, X_2^*, ..., X_T^*)$. It should be noted that in MDT, two implement methods is presented, one is to restore general deviation measure function and the other is to restore the derivative function of general deviation measure which admits time periods as input arguments. The correspondence relationship between the function arguments and time period is illustrated in the following figure.

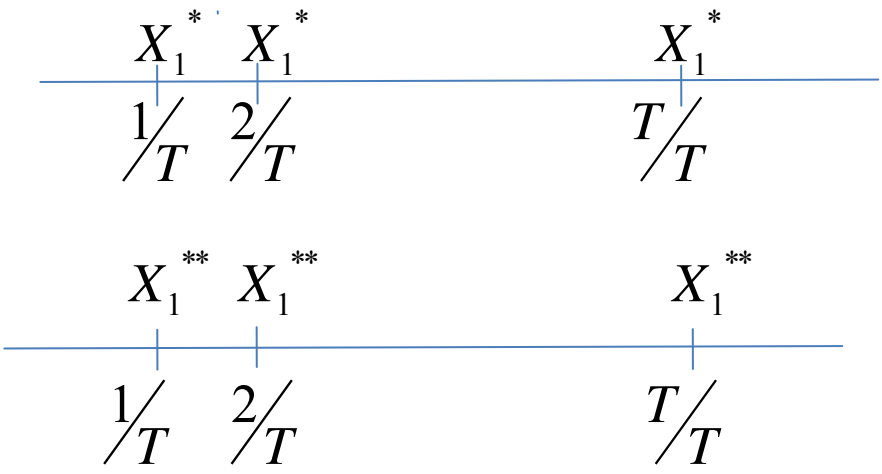


Figure_5

The reason for the show of above illustration is to highlight that different input arguments would be admitted by the restored derivative function of general deviation measure compared to the restored utility function and general deviation function.

In Figure_5, it should be pointed out that X* and X** are different random variables which denotes different choices of portfolio weights, and the two random variables should be correlated. However, in the implement of inverse portfolio problem, only a specified random variable X* is the concerns, which admits a set of random values over chosen number of time periods, i.e., $(X_1^*, X_2^*, ..., X_T^*)$, which are independent random variable measured from time perspective. The test of independence of these random observations will be presented in section 6 of the thesis.

2.4 The need to introduce SD and SSD

In actual financial markets, the most easily accessible information is the history trading prices such as stock prices. Based on this information, it is difficult to predict the future trade strategy. Using EUT and MDT, it becomes convenient to predict the proper trading strategy for individual investor because all prediction is based on the restored numerically approximate value of utility function and general deviation measure, which can be found using corresponding model.

However, the restoring process of the risk preference function is difficult to obtain through simple steps; that means, it is necessary to test whether there exists a numerical approximation of restored function before carrying out the modelling of the inverse portfolio problem. Therefore, it is necessary to introduce a relatively simple test method rather than waiting for the end of carrying out the restoring process of the inverse problem.

Then, the concept of stochastic dominance (SD) is introduced, which defines rational investment behaviour and also elaborates the relationship between stochastic dominance and concave function in the EUT and Schur-convex function in the MDT. Both SD and corresponding function represent concave ordering which denotes a risk preference between two choices of the portfolio weights of chosen number of assets. It should be noted that the risk preference should be considered to be a rational behaviour and can be a guide for individual investor.

In order to obtain better accuracy of risk preference decision, second-order stochastic dominance is adopted which can still be used to make a decision between two choices of portfolio weights in the case that expected value of two random variables takes equal value.

Besides, by introducing the concept of SD, it is more convenient to make a decision between two choices of the portfolio weights of chosen number of assets than traditional risk decision theory on the basis of proposed choice behaviour axioms.

2.5 The use of numerical restored result

The ultimate aim of the inverse portfolio problem is to generate the proper future trading strategy to the individual investor who holds a given portfolio value which happens to be the optimal portfolio value using restored utility theory and general deviation measure.

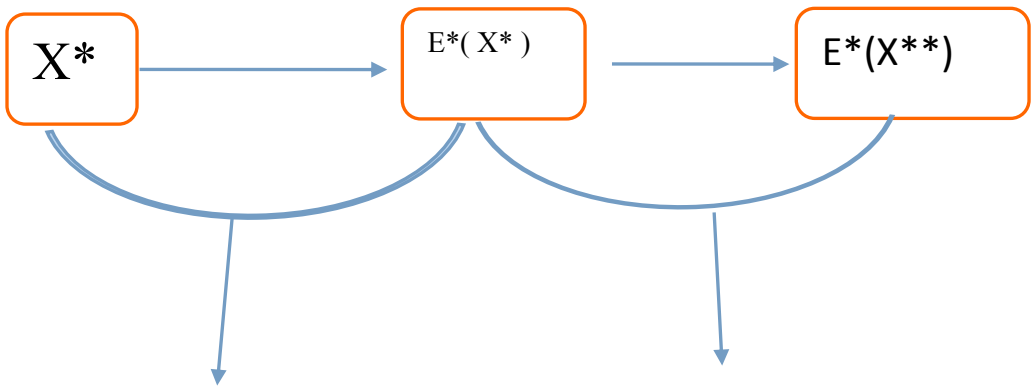
In the modelling of the inverse portfolio problem using EUT, the utility function is generated as a by-product when carrying on the second-order stochastic dominance (SSD) efficiency. It is constructed from slope coefficients vector (slope coefficient value on each time step) and intercept coefficients vector. If the investor would like to use the utility function to generate an optimal choice of portfolio weights, the portfolio optimization selection problem should be solved using the restored utility function. Although it is easy to model the problem and carry on the implement of the modelling, the whole project work has to be modelled two times, it is a bit non-trivial. However, using MDT, the future trade strategy and the restored deviation measure can be solved simultaneously using linear programming . In the thesis, we only do the modelling of the inverse problem. Therefore, in this section, the explanation of the use of the restored risk preference function, it is natural to choose the modelling example using MDT.

In the MDT, the project work can be divided into two steps, firstly, using a given portfolio value to generate a general deviation measure, after that, using the general deviation measure obtained to provide the individual investor with a trading strategy recommendation which is an altered portfolio weights vector based on the original the investor holds.

It should be noted that the recommended trading strategy, i.e. the alter portfolio weights vector is generated simultaneously with the restored general deviation measure., which is because the main idea of modelling using MDT is linear programming, and the optimal value and function value can be conveniently obtained as return values of optimization function.

Clear illustration of the project work looks like the following two figures. The first figure is using EUT and the second is in MDT (the restored function is general deviation measure rather than its derivative functions). In the figure, the assuming example in section (2.3) still applies, and the work starts from left to right.in order to distinguish with portfolio optimization selection problem, we take different notation D* rather than D in the MDT, similarly, take E* in EUT.

(1) Using EUT



The construction of utility function is based on slope coefficient vector and intercept coefficient vector.

The step is an optimization problem, it is a convex maximization.

Max $Eu(X)$

$X = \sum_{i=1}^{N} \lambda_i r_i$, then

$$u(X_j) = u(\sum_{i=1}^{N} \lambda_i r_{ij})$$

$$Eu(X) = \frac{1}{T} \sum_{j=1}^{T} u(\sum_{i=1}^{N} \lambda_i r_{ij})$$

Figure_6

(2) Using MDT , restore general deviation measure

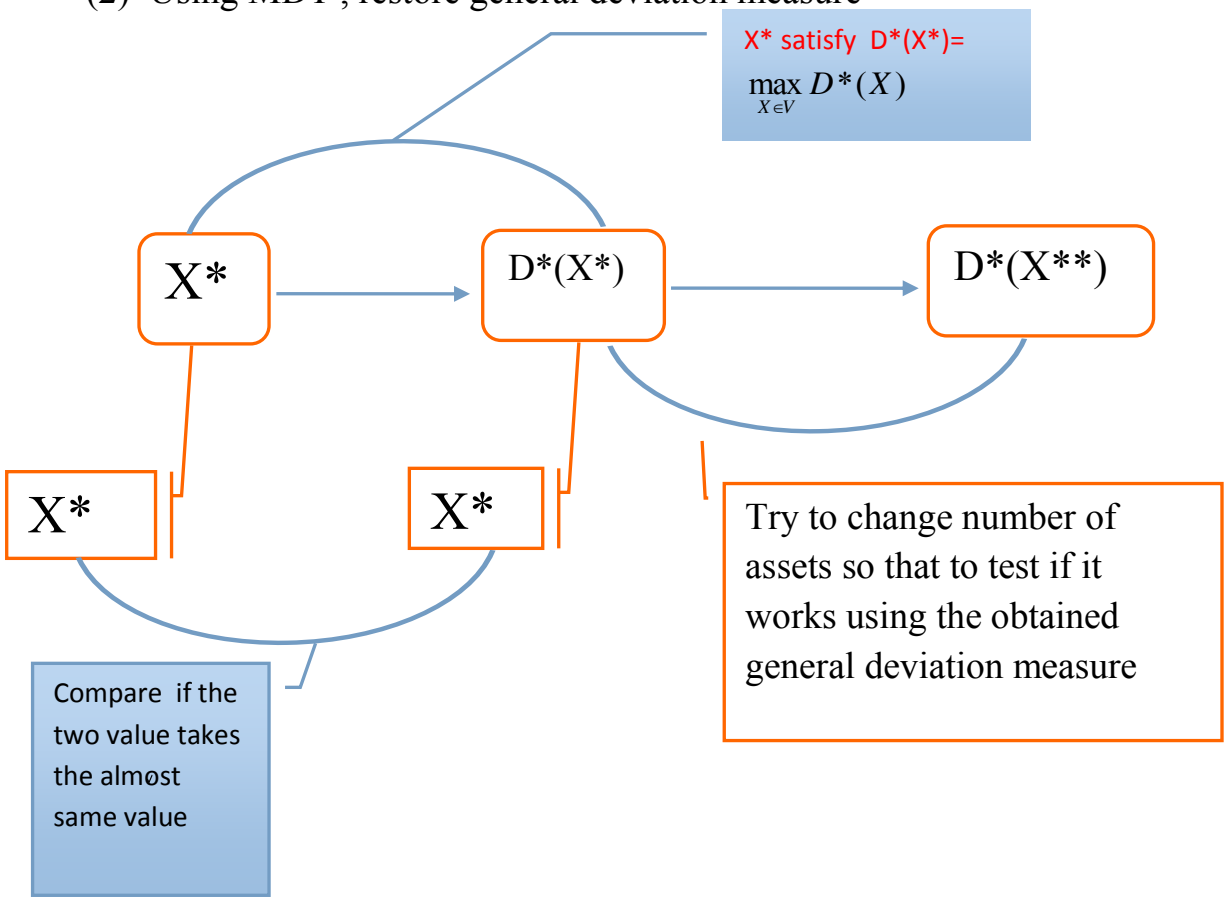


Figure_7

(3) Using MDT, restore the derivative function of general deviation measure

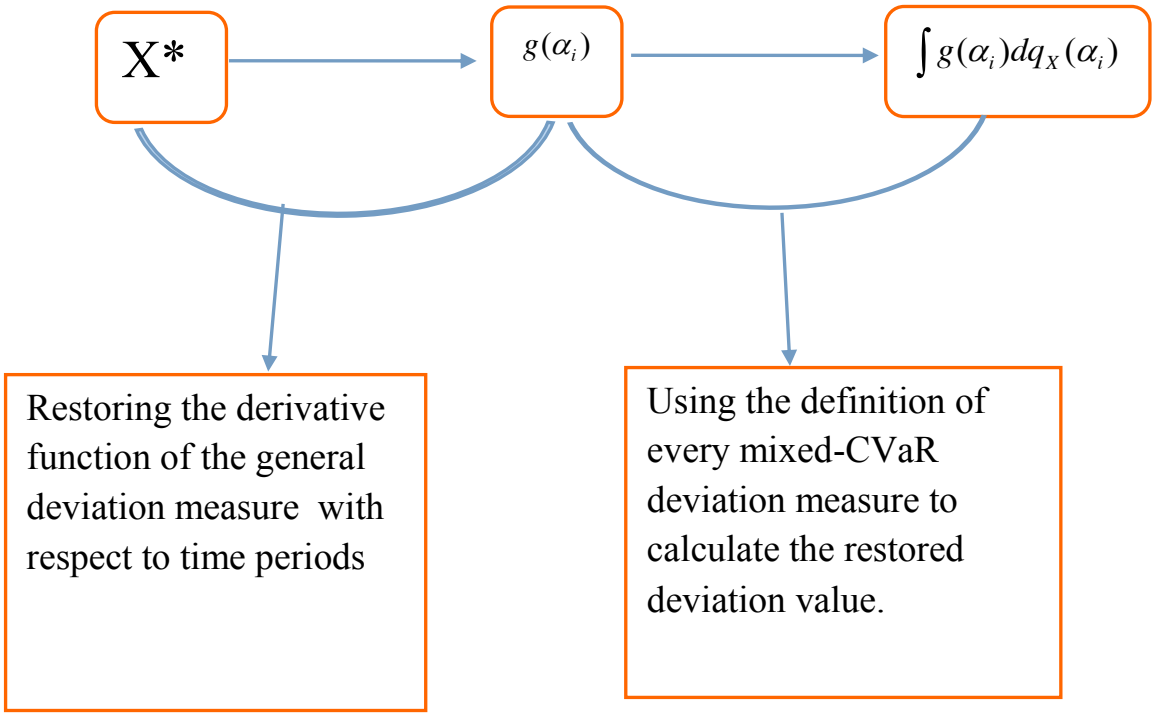


Figure_8

Chapter 3
Model Assumption

In this chapter, four main assumptions about modelling are presented. Firstly, the basic hypothesis about portfolio weights vector. Secondly, basic modelling context assumption is stated. Thirdly, the time series hypothesis is stated and random portfolio observations should follow independently identical distribution (IID). Fourthly, based on the modern portfolio theory (MPT) and the relation between SD and concave function in the EUT and Schur-convex function in the MDT, traditional concerns regarding optimization properties using these two theories are presented.

3.1 Assumption in terms of portfolio weights

The restoring process of inverse portfolio problem is begin with given optimal portfolio value, which is formulated from weight ratios of assets and is

correspondence with the portfolio weights vector. It is necessary to declare the assumption on these portfolio weights values, which should satisfying weight ration budget condition and the sum of all weight value equal to 1. Besides, the short sale is not allowed, that is , all portfolio weights value admits nonnegative value.

3.2 The modelling context assumption

In the thesis, the risk aversion case is only considered regarding risk tolerance preference of the individual investor.

3.3 Time series observations IID hypothesis

In the actual stock exchange, individual investors are exposed to a variety of different assets. In the restoring process of inverse portfolio problem, a specified optimal portfolio values, say, X* is the data concern. Measured from time perspective, the portfolio values based on chosen number of assets return is a random variable which admits a set of random values over chosen number of time periods, i.e. $(X_1^*, X_2^*, ..., X_T^*)$. From time perspective, the observations over time periods can be regarded as different random variables which is corresponded with different choice of portfolio weights. That is, each time period observations is a random variable. The property of these random variables is the object concern of time series IID test in the thesis.

According to central limit theorem (CLT), a large dimension of random observations would follow Gaussian distribution. The definition of CLT can be seen in [...]. It means that all random variables corresponding to all choices of portfolio weights over chosen time periods would follow the identical distribution. In the thesis, we omitted the time series Gaussian distribution test and only independency property of time series random variables is tested using three different method. Detailed test can be checked in section 6.

3.4 Traditional optimization property assumption in EUT and MDT

In the modern portfolio theory (MPT), the portfolio optimization selection problem is to maximizing the expected return of portfolio value for given amount of portfolio risk and to minimizing the risk for a given level of portfolio return. Specifically, in EUT, the portfolio optimization selection problem is to maximizing the expected utility return of portfolio value and to minimizing the general deviation measure function value on return of portfolio value using MDT. Based on the analysis above, together with the relationship

between SD and restored risk preference function in EUT and MDT respectively, the traditional optimization property can be deduced separately. That is, in the restoring process of inverse portfolio problem, if there exists several numbers of restored function, traditionally, the approach is to take the minimum value within the set of increasing utility concave function, say, U , and the set of Schur-convex function , say, W. It should be pointed out that general deviation measure is Schur-convex function.

Chapter 4
Notation Explanation

The explanations of mathematical notation in main five part of the modelling of restoring process of inverse portfolio problem is covered in this chapter. The main five part of modelling of the inverse problem is formulation of the portfolio return; portfolio optimization selection problem based on Markowitz Mean-Variance model; time series IID test model; modelling using EUT; modelling using MDT.

4.1 Notation for formulation of the portfolio return

$I \equiv [0,1,...N]$: index vector of the chosen number of assets;

$r_0, r_1, \ldots, r_N$: rate of returns of stock prick in the actual exchange; r_0 denotes risk-free or riskless interest rate, in realistic financial market, it is normally regarded as the interest rate of return on bank saving account. In the thesis, r_0 is set to equal to zero. That is, only risky assets return $r_1, \ldots, r_N$ is considered, i.e., the chosen number of assets is N;

$X = \sum_{i=0}^{N} \lambda_i r_i$: the portfolio weights ratio value vector;

: the return of portfolio return; in the thesis, as assuming $r_0 = 0$ The formulation of portfolio return only considers N numbers of assets. That is, $X = \sum_{i=1}^{N} \lambda_i r_i$

4.2 Notation of Markwitz portfolio optimization selection problem

μ : mean return of observation for each asset;

σ : standard deviation, i.e., the square root of variance;

$E[X] >= \pi$: expected return of portfolio is greater than the desired level of expected value. Generally, it takes the value greater or equal to the risk-free interest rate r_0 ; in Figure_2, assuming $\pi = r_0 = 0$

$\lambda^T \mu$: mean return of portfolio value;

$\sum$: covariance matrix;

$\lambda^T \sum \lambda$: the variance of portfolio return;

4.3 Notation of IID test model

τ **:** given choice of the portfolio weights;

$X^* \equiv (X_1^*, X_2^*, \ldots, X_T^*)$ **:** portfolio value over T time periods;

α**:** the possibility of maximum risk

$(y_1, \cdots, y_n)$ **:** a sequence of observations over n number of time periods, here, n denotes time;

T: number of turning points, difference signs, increasing pairs;

μ_T**:** mean value of random variable T;

σ_T**:** variance of random variable T;

$\Phi_{1-\alpha/2}$ **:** the $(1-\alpha/2)$ quantile of the standard normal distribution.

4.4 Notation of modelling using EUT

$[\beta_1, \beta_2, \ldots \beta_T]$: slope coeffecient vector on each time period;

$(\alpha_1, \ldots, \alpha_T)$: intercept vector on each time period;

4.5 Notation of modelling using MDT

$CVaR_\alpha^\Delta(X)$: conditional value-at-risk (CVaR) deviation, defined for any $\alpha \in (0,1)$ by $CVaR_\alpha^\Delta \equiv E(X) - \frac{1}{\alpha}\int_0^\alpha q_X(\beta)d\beta$

Mixed-CVaR deviation : $\int_0^1 CVaR_\alpha^\Delta(X)d\lambda(\alpha)$ Λ

Worst -case Mixed-CVaR deviation: $\sup_{\lambda\in\Lambda}\int_0^1 CVaR_\alpha^\Delta(X)d\lambda(\alpha)$, where denotes the set of choices of portfolio weights.

V: the set of portfolio value ;
W: Schur-convex function;
D*: supremum of all elements of W;

In the Worst-case Mixed-CVaR (try to restore derivatives of general deviation function)

$Q_i^* = \{q_{ik}^*\}_{k=1}^T$: the risk-envelope vector for each asset i over T time periods;

$\{c_{ij}\}_{i,j}$: coefficient matrix;

Result of Worst-case Mixed-CVaR:

α_i: the ith time period, equal to i/T;

g(α_i) : $g:(0,1)->R$, derivative function of general deviation measure

$q_X(\alpha)$: the risk-envelope of time period α , on given X

Chapter 5
Modelling

5.1 Mean_Variance_Model

5.1.1 Formulation(Mean-Variance Model)

- **Firstly, produce optimal portfolio weights using Markwitz Mean_Variance optimization, which will be used in the inverse problem.**
- **To Minimize the standard deviation of portfolio rate of return**
- **i.e. to find** X^* **s.t.** $\min_{X \in V} \sigma(X) = \sigma(X^*)$
- **form history prices, we know** $\mathrm{r} = (r_{1t}, \ldots, r_{nt})^{\mathrm{T}}$
- **then, mean return of observation of each period :** $\mu = \mathrm{E}(r_t)$
- **covariance matrix :** $\Sigma = \mathrm{Cov}(r_t)$
- **portfolio weights:** $\lambda = (\lambda_1, \ldots, \lambda_n)^{\mathrm{T}}$ **corresponding to variable** $\mathrm{X} \in \mathrm{V}$
- **The mean and the variance of the portfolio return:** $(\lambda^{\mathrm{T}}\mu, \lambda^{\mathrm{T}}\Sigma\lambda)$
- **Given** μ **, Find minimum variance** $\mathrm{w}^{\mathrm{T}}\Sigma\mathrm{w}$ **equal to find** $(1/2)*\mathrm{w}^{\mathrm{T}}\Sigma\mathrm{w}$
- **Therefore, Mean_Variance optimization can use quadratic programming**

5.1.2 Generate Optimal Portfolio

- **Step1: quadratic programming using Markwitz Mean_Variance model ;**
- **Step2: bar graph of the optimal portfolio weights ;**
- **Step3: produce portfolio values using the weights over T periods;**

5.2 Modelling of relation between SD and expected restored function

5.2.1SSD Defination

- **An r.v. X dominates Y with respect to second-order stochastic dominance (SSD), i.e.** $X \succeq_{SSD} Y$ **if** $E[f(X)] \geq E[f(Y)]$ **for every concave increasing function** $f: R \to R$

5.2.2 SSD-efficient Definition

- **An r.v.** X^* **is SSD inefficient if and only if some** $X \succeq_{SSD} X^*$ **;** $*$

- **Alternatively, X^* is SSD efficient if and only if no r.v. X s.t. $X \succeq_{SSD} X^*$;**
- **from $*$ we can derive that : X^* is SSD inefficient if and only if, for all utility functions $u \in U$, $E[u(X)] \geq E[u(X^*)]$, that is we would not find u s. t. the given portfolio return X^* is optimal for the inversion problem.**
- **alternatively, SSD inefficient means for all u , $\max_{X \in V} E[u(X)] \geq E[u(X^*)]$, i.e. $\min_{u \in U}\left(\max_{X \in V} E[u(X)] \geq E[u(X^*)]\right) = 0$**
- **X^* is SSD efficient means means for some u ,**

  $\min_{u \in U}\left(\max_{X \in V} E[u(X)] \geq E[u(X^*)]\right) = 0$ Then, SSD efficient can be rephased in terms of minimax formulation;

5.2.3 Relation with Formulation in EUT

In EUT, if there exists several number of restored utility function, it is traditionally to take minimum value within the set of concave utility functions, say, U.

5.2.4 Relation with Formulation in MDT

In MDT, if there exists several number of restored general deviation measure, it is traditionally to take maximum value within the set of Schur-convex function, say, W.

5.3 Expected_Utility_Model

5.3.1 Formulation for SSD efficient test

- **from linear programming get a vector: $[\beta_1, \beta_2, \ldots \beta_T, \theta]$**
- **For each asset i , we only need $[\beta_1, \beta_2, \ldots \beta_T]$ to construct utility function, which is formulated as $\min_{t \in \Theta}(\alpha_t + \beta_t x)$**
- **$[\beta_1, \beta_2, \ldots \beta_T]$ be called slope coeffecient, and $\beta_1 > \beta_2 > \ldots > \beta_T = 1$**
- **As the relation $\alpha_t + \beta_t x_t = \alpha_{t+1} + \beta_{t+1} x_t, t \in \Theta$, and starting from $\alpha_T = 0$,**

the intercept coefficient $(\alpha_1, \ldots, \alpha_T)$ can be formulated as $\alpha_t = \sum_{s=t}^{T-1} (\beta_{s+1} - \beta_s) x_s, t \in [1, \ldots T-1]$, it is a piecewise-linear function

- **Then constructing and graph utility function for each asset i , the**

The figure we get will be consistent with theoretical utility graph.

5.3.2 Implement

For Utility function for N=3,T=4

- **make sure**
- **for asset i=1, i=2 ,i=3**

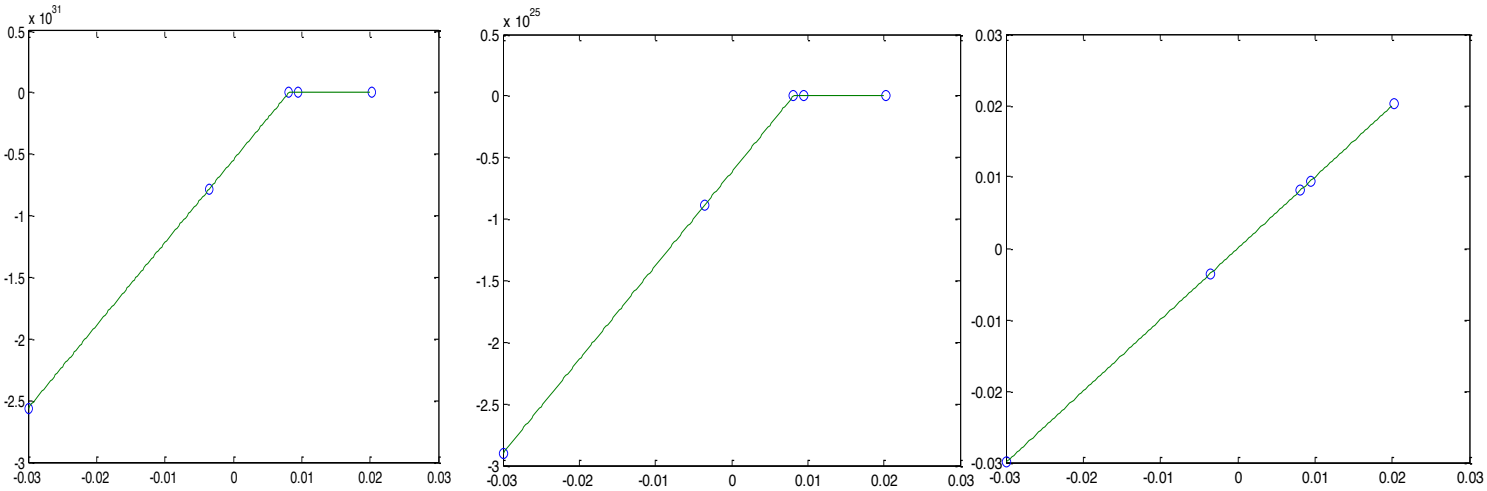


Figure_9

5.4 Mean_Deviation_Model

5.4.1 Two propositions

- **Introduce two methods : firstly, the worst-case mixed_CVaR deviation ;**
- **Then, mixed_CVaR deviation (get unique deviation measure)**
- **Proposition 1 : given V and X* be** $\succeq_S undo\min ated$ **, then there exist nonnegative weights** $\lambda_1,\dots,\lambda_{T-1}$ **with** $\sum_{i=1}^{N}\lambda_i = 1$ **s.t X* is a solution of inverse problem for the deviation** $D(X)=\sum_{i=0}^{T-1}\lambda_i CVaR^{\Delta}_{i/T}(X)$ **;**

- **Proposition 2: If non-empty , the set W contains the unique ‘‘maximal’’ element, i.e. there exists a unique deviation measure** $D^* \in w$ **s.t.** $D^*(X) \geq D(X)$

for all $X \in V$ **and** $D \in w$ **,In fact,** D^* **is the supremum of all elements of W :**

$D^*(X) = \pi \sup_{\alpha \in [0,1]} D(X)$ **for all** $X \in V$ **, equavalently, it admits :**

$$\boxed{D^*(X) = \pi \sup_{\alpha \in [0,1]} \frac{CVaR_\alpha^\Delta(X)}{CVaR_\alpha^\Delta(X^*)}}$$

5.4.2 Formulation For Worst_case Mixed_CVaR

- **Firstly,** $Q_i^* = \{q_{ik}^*\}_{k=1}^T$ **using formula** $\min_{q_{ik}} \sum_{i=0}^{n} q_{ik} X_k$ **s.t.** $\sum_{k=1} q_{ik} = T$ **,** $0 \leq q_{ik} \geq 1/\alpha_i$ **,** $k = 1, \ldots, K; K = T$
- **Secondly, find weights vector** $(\lambda_1, \ldots, \lambda_{T-1})$ **through solving linear**

equation $\sum_{i=1}^{T-1} c_{ij}\lambda_i = 0, j = 1, \ldots n$ **with**

$c_{ij} = (E(r_j) - r_0) CVaR_{\alpha_i}^\Delta(X^*) + (E(X^*) - r_0) \operatorname{cov} ar(r_j, Q_i^*)$

for i=1, …, T-1; j=1, … ,n

and with $\sum_{i=0}^{n} \lambda_i = 1, \lambda_i \geq 0, i = 1, \ldots, T-1$

- **for the second step, as we need to find constrained solution of the**

system of linear equation, so, rather than use ‘linsolve’, we change to

use ‘lsqlin’ to solve least-squares curve fitting problems. The solution

is the same.

- **the reason to perform this change method, because we try to use**
- **the solution : weight vector to solve** $\boxed{\min_{\lambda_1, \ldots \lambda_{T-1}} \sum_{j=1}^{n} (\sum_{i=1}^{T-1} c_{ij}\lambda_i)^2 w^*, j = 1, \ldots n}$
- **we know the sum of given optimal portfolio weights equal to 1, it means**

to solve $\boxed{\min_{\lambda_1, \ldots \lambda_{T-1}} \sum_{j=1}^{n} (\sum_{i=1}^{T-1} c_{ij}\lambda_i)^2, j = 1, \ldots n}$**, so that we can use ‘lsqlin’**

5.4.3 Implement of the worst-case mixed CVaR

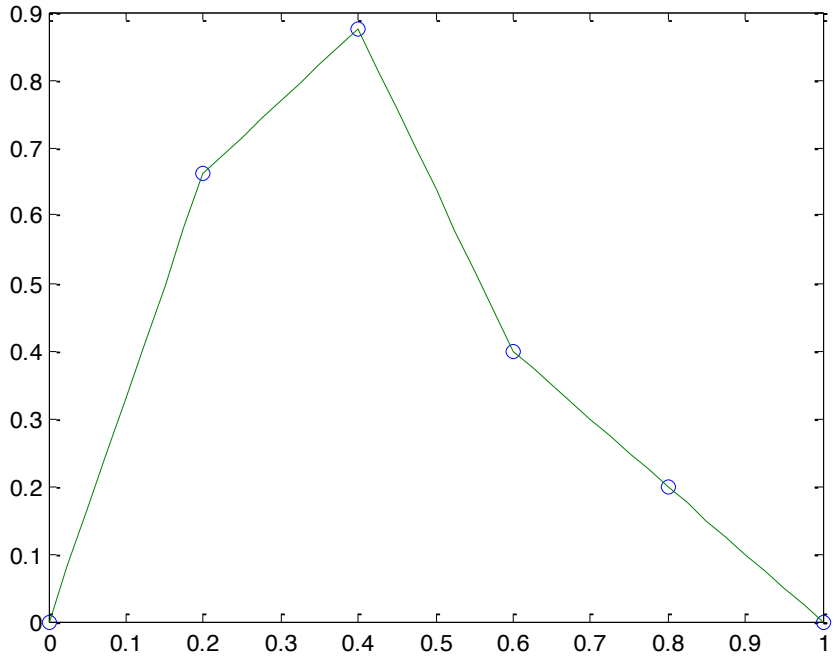
- **in this method, we will get a number of function g(x) ,** $x=\alpha(i)=i/T, i=1,...,T-1$ **,**

then, use the definition for every Mixed_CVaR deviation: $D(X)=\int_0^1 g(\alpha)d(q_x(\alpha))$ **, for** $g:(0,1)->R$ **some positive concave function .**

- **step1: minimum optimization find** Q_i* **using given portfolio value X*;**
- **step2: generate weights** $\lambda_1,\dots,\lambda_{T-1}$ **using optimality conditions for inverse problem with the worst_case mixed CVaR deviation** $\boxed{D(X)=\sum_{i=0}^{T-1}\lambda_i CVaR^{\Delta}_{i/T}(X)}$
- **step3: minimum optimization find** $\boxed{\min_{x\in V} D(X)=\sum_{i=0}^{T-1}\lambda_i CVaR^{\Delta}_{i/T}(X)}$ **or alternatively, using the solution found in step2 and defination** $\int_0^1 g(\alpha)d(q_x(\alpha))$ **to get D(X).**
- **The disadvantage: in the result of step3, it is possible to get more than one deviation measure , so omit step3, we just finish step2.**

5.4.4 The restored graph

- **For N=10, T=11, it is consistent with expected graph.**



5.4.5 Suggestion to Improve Mixed_CVaR deviation(unique)

$$\boxed{D^*(X) = \pi \sup_{\alpha \in [0,1]} \frac{CVaR_\alpha^\Delta(X)}{CVaR_\alpha^\Delta(X^*)}}$$

in the discrete model, the formulation admits:

$$\boxed{\begin{aligned} &\min_{X \in V, u, c_i, Z_i} u \\ &EX - c_i + (T/i)EZ_i \le u\, CVaR_{\alpha_i}^\Delta(X^*), \\ &Z_i \ge c_i - X, \\ &Z_i \ge 0, i = 1, \ldots, T-1 \\ &EX \ge \pi \end{aligned}}$$

It should be noted that in the above formulation condition, Z_i IS a vector.

Chapter 6
Time Series Observation Hypothesis Test

In this chapter, three testing method is introduced. The steps is as follows.

- we based assumption that the timing of drawings is inconsequential, so that in the computation process, the time-series observation can be ranked with respect to the given portfolio $\boxed{X^*}$, i.e. $x_1^T\tau < x_2^T\tau < \ldots < x_t^T\tau$
- Therefore, we need to carry on the time independence test.
- IID TEST (use $\alpha = 0.05$), the test result should return logical value TRUE.
- here, introduce three types of iid test for a sequence of observations

- $y_1, \ldots, y_n$, 1<i<n, (n denotes time) . We test if $|T-\mu_T|/\sigma_T > \Phi_{1-\alpha/2}$
- $\Phi_{1-\alpha/2}$ is the $(1-\alpha/2)$ quantile of the standard normal distribution.
- 1. turning points test (if $y_{i-1} < y_i$ and $y_i > y_{i+1}$ Or if $y_{i-1} > y_i$ and $y_i < y_{i+1}$)
- 2. difference sign test (if $y_i > y_{i-1}$, $i = 2, \ldots, n$)
- 3. rank test (the number of pairs (i, j) s.t. $y_j > y_i$
  And $j > i$, $i = 1, \ldots, n-1$)
- All test function outputs the logical value, if equal to 1, it means, the iid
- hypothesis is accepted, otherwise , it is rejected .

Chapter 7
Model Test

The restored risk preference using EUT and MDT should be consistent with Figure_3, and the followed figure in section 1.3.

Chapter 8
Conclusion

The ultimate aim of the project is to use the numerical restored risk preference function to recommend an optimal choice of portfolio weights to the investor based the portfolio weights vector he or she holds originally. The detailed explanation can be checked in problem analysis part of the thesis.

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- http://homepages.math.uic.edu/~jan/mcs320s08/matlec9.pdf
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Appendix

Appendix pre-fix: user guide for matlab program

(1). Firstly, produce optimal portfolio weights using Markwitz Mean_Variance optimization ; Then, use the optimal weights to get given portfolio values X*

```
>> N=10,T=11

>> x1=optiml_portfolio_mean_varian( N,T )

>> plotPortfDemoStandardModel(x1)

>> portfolio_given(N,T)
```

(2). Implement restoring process in EUT, choosing N=3,T=4

```
>> portfolio_given(  )

>> beta(3,4)

>> alpha(3,4)

>> utility(1,3,4)

>> utility(2,3,4)

>> utility(3,3,4)
```

(3). IID TEST: (All test function outputs the logical value, if equal to 1, that means, the iid hypothesis is accepted;; otherwise , it is rejected)

```
(1). // turning_point_test.m

>> N=3,T=4,alpha=0.05

>> turning_point_test( N,T,alpha )

(2). difference_sign_test.m

>> difference_sign_test( N,T,alpha )

(3). // rank_test.m

>> rank_test( N,T,alpha )
```

Result: All three testing method can test the given optimal choice of portfolio weights follows IID.

(4) Implement restoring process in MDT, to restore derivative function of general deviation measure.

Step1. riskI_envelope.m;; [optimization problem for asset i]

Step2. mean_returnOfobservation.m;;

[obtain mean return based on observation over T time periods]

1. CVaR_asset_I.m [CVaR number for asset i;;]
2. Covar_asset_I.m [covar number for asset I;; it is not covariance, it is the average value evaluation based on risk envelop for each asset i ;;]
3. Mix_CVaR_coeff.m [for worst_case Mixed CVaR]

CVaR_optimality_condition.m

4. risk_profile

   Test:

   >> risk_profile(N,T)

   The expected should look like a semi-circle in 1.3.

(5) Implement restoring process in MDT, to restore uniquely the general deviation measure.

Evaluate I=1, for 1/T, the deviation measure

```
>> I=1

>> deviation_measure_unique( I,N,T )
```

Appendix I

1.// turning_point_test.m

```
function [ LV ] = turning_point_test( N,T,alpha )
%TURNING_POINT_TEST Summary of this function goes here
%   the output parameter is a logical value;;

P = [0.0162355710000000,-0.00246812000000000,-0.0404016810000000,-
0.0419082760000000,-0.0737995490000000,-0.0340909090000000,-
0.0419082760000000,1645.88427300000,0.0354098360000000,-0.184210526000000;
    0.0257420960000000,-
0.00515463900000000,0.0194694570000000,0.0707015130000000,-
0.00939457200000000,0.0745098040000000,0.0707015130000000,0.171171171000000
,0.0232742240000000,-0.0645161290000000;
    0.0265704650000000,0.0130569950000000,0.00381952700000000,-
0.00539568300000000,-0.0386371620000000,-0.0364963500000000,-
0.00539568300000000,0.107692308000000,0.0479653410000000,0.0689655170000000;
    0.0131094440000000,-
0.0180032730000000,0.0623067780000000,0.0211831570000000,0.0372670810000000
,0.121212121000000,0.0211831570000000,0.00694444400000000,0.022442049000000
0,0.322580645000000;
    -
0.0114467550000000,0.0729166670000000,0.0315648090000000,0.0131545660000000
,-0.126805213000000,0.0405405410000000,0.0131545660000000,-
0.999508966000000,-0.0127075810000000,0.121951220000000;
    0.0142033320000000,0.106796117000000,-0.0108506940000000,-
0.00549313400000000,-0.0338846310000000,0.0324675320000000,-
0.00549313400000000,1515.85393300000,0.0783969580000000,-0.0652173910000000;
-0.00970231200000000,-0.0417543860000000,-0.0217200530000000,-
0.0271152400000000,0.0237995820000000,-0.0283018870000000,-
0.0271152400000000,-0.999390741000000,0.0187169400000000,-
0.0465116280000000;
    -0.0126682260000000,0.0252654710000000,-0.0461986990000000,-
0.0237419350000000,0.154159869000000,0.0226537220000000,-
0.0237419350000000,-0.0516717330000000,0.0171748100000000,-
0.0731707320000000;
    -
0.00121538500000000,0.0446428570000000,0.00681871600000000,0.00211472400000
000,-0.0155477030000000,0.107594937000000,0.00211472400000000,-
0.0160256410000000,-0.0210732980000000,-0.0263157890000000;
    -0.00787110500000000,0.0769230770000000,-0.0366651100000000,-
0.0308625690000000,-0.113424264000000,-0.00571428600000000,-
0.0308625690000000,-0.0358306190000000,-0.0299505280000000,-
0.0540540540000000;

0.00523210700000000,0.178571429000000,0.0729696970000000,0.0204137180000000
,0.0700404860000000,0.0890804600000000,0.0204137180000000,488.864864900000,
0.00110268800000000,0;];
P=P'; % P: (N-by-T)
r=P';
lamda = optiml_portfolio_mean_varian(N,T);
```

```
lamda=lamda';
ux=lamda*r(1,:)';
for t=2:4
  ux=[ux,lamda*P(:,t)];
end
%%%%%
 d1=sign(diff(ux));
 d2=diff(d1);
 num_turn_point=numel(find(abs(d2))==2);
%%%
 crite=norminv(1-alpha/2);
 drift=(2/3)*(T-2);
 variance=(16*T-29)/90;
 std=sqrt(variance);
 if abs(num_turn_point-drift)/std > crite
    LV=false(ones);
 else
    LV=true(ones);
 end
end
```

2.// difference_sign_test.m

```
function [ LV ] = difference_sign_test( N,T,alpha )
%DIFFERENCE_SIGN_TEST Summary of this function goes here
%   Detailed explanation goes here
P = [0.013109444000000  -0.011446755000000   0.014203332000000  -
0.006573504000000;
  -0.006901994000000  -0.016561345000000   0.021431869000000
0.004131142000000;
   0.012556248000000   0.051666835000000   0.026383835000000  -
0.005190110000000];
r=P';
lamda = optiml_portfolio_mean_varian(N,T);
lamda=lamda';
ux=lamda*r(1,:)';
for t=2:4
  ux=[ux,lamda*P(:,t)];
end
%%%
d=diff(ux);
num_diff_positv=numel(find(sign(d)==1));
%%%
crite=norminv(1-alpha/2);
drift=(T-1)/2;
variance=(T+1)/12;
std=sqrt(variance);
if abs(num_diff_positv-drift)/std > crite
  LV=false(ones);
else
  LV=true(ones);
end
end
```

3.// rank_test.m

```
function [ LV ] = rank_test( N,T,alpha )
```

```
%RANK_TEST Summary of this function goes here
%   Detailed explanation goes here
P = [0.013109444000000  -0.011446755000000   0.014203332000000  -
0.006573504000000;
  -0.006901994000000  -0.016561345000000   0.021431869000000
0.004131142000000;
   0.012556248000000   0.051666835000000   0.026383835000000  -
0.005190110000000];
r=P';
lamda = optiml_portfolio_mean_varian(N,T);
lamda=lamda';
ux=lamda*r(1,:)';
for t=2:4
  ux=[ux,lamda*P(:,t)];
end
%%%
num_pair_positv=0;
for j=T:-1:2
   dj=ux(j)-ux(1:j-1);
   num_pair_positv=num_pair_positv+numel(find(dj>0));
end
%%%
crite=norminv(1-alpha/2);
drift=T*(T-1)/4;
variance=T*(T-1)*(2*T+5)/8;
std=sqrt(variance);
if abs(num_pair_positv-drift)/std > crite
   LV=false(ones);
else
   LV=true(ones);
end
end
```

Appendix II

1.//beta.m

```
function [ XX ] = beta( N,T )
%BETA Summary of this function goes here
P = [0.013109444000000  -0.011446755000000   0.014203332000000  -
0.006573504000000;
  -0.006901994000000  -0.016561345000000   0.021431869000000
0.004131142000000;
   0.012556248000000   0.051666835000000   0.026383835000000  -
0.005190110000000];
lamda = optiml_portfolio_mean_varian(N,T);
lamda=lamda';
ux=lamda*P(:,1);
for t=2:T
  ux=[ux,lamda*P(:,t)];
end
ux=sort(ux);
 f=zeros(1,T+1);
 f(1,T+1)=1;
 bet=zeros(N,T+1);
 X=zeros(N,T+1);     %the solution of linear programming
 XX=zeros(N,T);      % the solution of beta
 B=zeros(N,T);
 C=zeros(N,T);
```

```
for i=1:N
    for t=1:T
         B(i,t)=  P(i,t)-ux(t);
    end
end
for i=1:N
       uB=B(i,:);
       C(i,:)=uB(T:-1:1);
end
 e1=ones(N,1);
 e2=ones(T-1,1);
 ub1=spdiags([-1*e2],1,T-1,T);
 ub2=spdiags([e2,-1*e2],0:1,T-1,T);
 z=zeros(T-1,1);
 ub1=[ub1,z];
 ub2=[ub2,z];
%%%%%%%%
 bet=[C,-1*e1];
 for i=1:N
    A=[bet(i,:);ub1;ub2];
    b=zeros(size(A,1),1);
    Aeq=zeros(1,T+1);
    Aeq(1,1)=1;
    beq=ones(size(Aeq,1),1);
    X(i,:)=linprog(f,A,b,Aeq,beq);
    tmx=X(i,:);
    XX(i,:)=tmx(T:-1:1);
 end
 end
```

2.//alpha.m

```
function [ alph ] = alpha( N,T)
%ALPHA Summary of this function goes here
ux=portfolio_given(N,T);
alph=zeros(N,T);
sum=zeros(N,T-1);
xx=beta(N,T);
for i=1:N
  for s=1:(T-1)
     for j=s:(T-1)
         sum(i,s)=(xx(i,j+1)-xx(i,j))*ux(j)+sum(i,s);
     end
  end
end
alph_T=zeros(N,1);
sum=[sum,alph_T];
alph=sum;
end
```

3.//utility.m

```
function [ output_args ] = utility( I,N,T )
%UTILITY Summary of this function goes here
% I denotes the asset I
ux=portfolio_given(N,T);
alph=zeros(N,T);
bet=zeros(N,T);
alph=alpha(N,T);
```

```
bet=beta(N,T);

%%%%%
y1={0,0,0,0};
z1=zeros(N,T);
for i=2:3
    y1=[y1;{0,0,0,0}];
end
for i=1:N
  for t=1:T
    y1{i,t}=@(x)bet(i,t)*x+alph(i,t);
    z1(i,t)=y1{i,t}(ux(t));
  end
end
uxx=[-0.03,ux];
for i=1:N
   zz1(i)=y1{i,1}(-0.03);
end
zz=[zz1',z1];
uxi=uxx(1):0.00025:uxx(T+1);
uyi = interp1(uxx,zz(I,:),uxi);
plot(uxx,zz(I,:),'O',uxi,uyi);
end
```

Appendix III

1.// optiml_portfolio_mean_varian.m

```
function [ x1 ] = optiml_portfolio_mean_varian( N,T )
%OPTIML_PORTFOLIO_MEAN_VARIAN Summary of this function goes here
%N=10,T=11,P:(T-by-N)
P = [0.0162355710000000,-0.00246812000000000,-0.0404016810000000,-
0.0419082760000000,-0.0737995490000000,-0.0340909090000000,-
0.0419082760000000,1645.88427300000,0.0354098360000000,-0.184210526000000;
    0.0257420960000000,-
0.00515463900000000,0.0194694570000000,0.0707015130000000,-
0.00939457200000000,0.0745098040000000,0.0707015130000000,0.171171171000000
,0.0232742240000000,-0.0645161290000000;
    0.0265704650000000,0.0130569950000000,0.00381952700000000,-
0.00539568300000000,-0.0386371620000000,-0.0364963500000000,-
0.00539568300000000,0.107692308000000,0.0479653410000000,0.0689655170000000;
    0.0131094440000000,-
0.0180032730000000,0.0623067780000000,0.0211831570000000,0.0372670810000000
,0.121212121000000,0.0211831570000000,0.00694444400000000,0.022442049000000
0,0.322580645000000;
    -
0.0114467550000000,0.0729166670000000,0.0315648090000000,0.0131545660000000
,-0.126805213000000,0.0405405410000000,0.0131545660000000,-
0.999508966000000,-0.0127075810000000,0.121951220000000;
    0.0142033320000000,0.106796117000000,-0.0108506940000000,-
0.00549313400000000,-0.0338846310000000,0.0324675320000000,-
0.00549313400000000,1515.85393300000,0.0783969580000000,-0.0652173910000000;
-0.00970231200000000,-0.0417543860000000,-0.0217200530000000,-
0.0271152400000000,0.0237995820000000,-0.0283018870000000,-
0.0271152400000000,-0.999390741000000,0.0187169400000000,-
0.0465116280000000;
    -0.0126682260000000,0.0252654710000000,-0.0461986990000000,-
0.0237419350000000,0.154159869000000,0.0226537220000000,-
```

```
0.0237419350000000,-0.0516717330000000,0.0171748100000000,-
0.0731707320000000;
    -
0.00121538500000000,0.0446428570000000,0.00681871600000000,0.00211472400000
000,-0.0155477030000000,0.107594937000000,0.00211472400000000,-
0.0160256410000000,-0.0210732980000000,-0.0263157890000000;
    -0.00787110500000000,0.0769230770000000,-0.0366651100000000,-
0.0308625690000000,-0.113424264000000,-0.00571428600000000,-
0.0308625690000000,-0.0358306190000000,-0.0299505280000000,-
0.0540540540000000;

0.00523210700000000,0.178571429000000,0.0729696970000000,0.0204137180000000
,0.0700404860000000,0.0890804600000000,0.0204137180000000,488.864864900000,
0.00110268800000000,0;];
P=P';  %%P: (N-by-T)
r=P';
mean_return=mean_returnOfobservation(N);  %asset I
%alternatively, use command mean(r), get the same result...

%%desired return;; % risk_free fixed saving account interest rate
%%choose fixed saving rate 2.03% for one year, we need weekly rate;;
rfm=0.0203;    % interest rate for one year
rfm=(rfm+1)^(7/365)-1 ; % weekly risk_free interst rate chosen
%%%
stdDev_return=std(r,1,1);  %%% set flag=1 (the second input parameter)
correlation=corrcoef(r);
a1=stdDev_return'*stdDev_return;
covariance=correlation.*a1;
nAssets=numel(mean_return);

Aeq=ones(1,nAssets);
beq=1;
Aineq=-mean_return;
bineq=-rfm;
lb=zeros(nAssets,1);
ub=ones(nAssets,1);
c=zeros(nAssets,1);      %%%objective function has no linear term
options=optimset('quadprog');    %%%default options for the solver
quadprog;;
options = optimset(options,'Algorithm', 'interior-point-convex');
options = optimset(options,'Display','iter','TolFun',1e-10); %%%additional
option setting
tic
[x1,fval1]=quadprog(covariance,c,Aineq,bineq,Aeq,beq,lb,ub,[],options);
toc
end
```

2.// plotPortfDemoStandardModel.m

```
function plotPortfDemoStandardModel(x1)
% plotPortfDemoStandardModel Helper function for portfolio optimization
demo

figure;
bar1 = bar(x1,'FaceColor','b','EdgeColor','b');
set(bar1,'BarWidth',0.2);
%set(gca,'xlim',[1 length(x1)])
%set(gca,'ylim',[0 0.3])
```

```
%set(gca,'xTick',[1 75 150 225]);
title('Mean-Variance-Standard model - 10-asset problem')
xlabel('Assets')
ylabel('Fraction of investment')
grid on
end
```

3.// portfolio_given.m

```
function [ ux ] = portfolio_given(N,T)
%PORTFOLIO_GIVEN Summary of this function goes here
%%N=10,T=11,P:(T-by-N)
P = [0.0162355710000000,-0.00246812000000000,-0.0404016810000000,-
0.0419082760000000,-0.0737995490000000,-0.0340909090000000,-
0.0419082760000000,1645.88427300000,0.0354098360000000,-0.184210526000000;
    0.0257420960000000,-
0.00515463900000000,0.0194694570000000,0.0707015130000000,-
0.00939457200000000,0.0745098040000000,0.0707015130000000,0.171171171000000
,0.0232742240000000,-0.0645161290000000;
    0.0265704650000000,0.0130569950000000,0.00381952700000000,-
0.00539568300000000,-0.0386371620000000,-0.0364963500000000,-
0.00539568300000000,0.107692308000000,0.0479653410000000,0.0689655170000000;
    0.0131094440000000,-
0.0180032730000000,0.0623067780000000,0.0211831570000000,0.0372670810000000
,0.121212121000000,0.0211831570000000,0.00694444400000000,0.022442049000000
0,0.322580645000000;
    -
0.0114467550000000,0.0729166670000000,0.0315648090000000,0.0131545660000000
,-0.126805213000000,0.0405405410000000,0.0131545660000000,-
0.999508966000000,-0.0127075810000000,0.121951220000000;
    0.0142033320000000,0.106796117000000,-0.0108506940000000,-
0.00549313400000000,-0.0338846310000000,0.0324675320000000,-
0.00549313400000000,1515.85393300000,0.0783969580000000,-0.0652173910000000;
-0.00970231200000000,-0.0417543860000000,-0.0217200530000000,-
0.0271152400000000,0.0237995820000000,-0.0283018870000000,-
0.0271152400000000,-0.999390741000000,0.0187169400000000,-
0.0465116280000000;
    -0.0126682260000000,0.0252654710000000,-0.0461986990000000,-
0.0237419350000000,0.154159869000000,0.0226537220000000,-
0.0237419350000000,-0.0516717330000000,0.0171748100000000,-
0.0731707320000000;
    -
0.00121538500000000,0.0446428570000000,0.00681871600000000,0.00211472400000
000,-0.0155477030000000,0.107594937000000,0.00211472400000000,-
0.0160256410000000,-0.0210732980000000,-0.0263157890000000;
    -0.00787110500000000,0.0769230770000000,-0.0366651100000000,-
0.0308625690000000,-0.113424264000000,-0.00571428600000000,-
0.0308625690000000,-0.0358306190000000,-0.0299505280000000,-
0.0540540540000000;

0.00523210700000000,0.178571429000000,0.0729696970000000,0.0204137180000000
,0.0700404860000000,0.0890804600000000,0.0204137180000000,488.864864900000,
0.00110268800000000,0;];
P=P'; % P: (N-by-T)
%%lamda =[0.410000000000000   0.240000000000000 0.350000000000000];
lamda = optiml_portfolio_mean_varian(N,T); %given optimal portfolio weights
lamda=lamda';
ux=lamda*P(:,1);
for t=2:T
```

```
  ux=[ux,lamda*P(:,t)];
end
ux=sort(ux); % rank portfolio observations due to the timing of draws are
inconsequential
end
```

4.// Covar_asset_I.m

```
function [ cova ] = Covar_asset_I( I,J,N,T )
%COVAR_ASSET_I Summary of this function goes here
%   Detailed explanation goes here
P = [0.0162355710000000,-0.00246812000000000,-0.0404016810000000,-
0.0419082760000000,-0.0737995490000000,-0.0340909090000000,-
0.0419082760000000,1645.88427300000,0.0354098360000000,-0.184210526000000;
    0.0257420960000000,-
0.00515463900000000,0.0194694570000000,0.0707015130000000,-
0.00939457200000000,0.0745098040000000,0.0707015130000000,0.171171171000000
,0.0232742240000000,-0.0645161290000000;
    0.0265704650000000,0.0130569950000000,0.00381952700000000,-
0.00539568300000000,-0.0386371620000000,-0.0364963500000000,-
0.00539568300000000,0.107692308000000,0.0479653410000000,0.0689655170000000;
    0.0131094440000000,-
0.0180032730000000,0.0623067780000000,0.0211831570000000,0.0372670810000000
,0.121212121000000,0.0211831570000000,0.00694444400000000,0.022442049000000
0,0.322580645000000;
    -
0.0114467550000000,0.0729166670000000,0.0315648090000000,0.0131545660000000
,-0.126805213000000,0.0405405410000000,0.0131545660000000,-
0.999508966000000,-0.0127075810000000,0.121951220000000;
    0.0142033320000000,0.106796117000000,-0.0108506940000000,-
0.00549313400000000,-0.0338846310000000,0.0324675320000000,-
0.00549313400000000,1515.85393300000,0.0783969580000000,-0.0652173910000000;
-0.00970231200000000,-0.0417543860000000,-0.0217200530000000,-
0.0271152400000000,0.0237995820000000,-0.0283018870000000,-
0.0271152400000000,-0.999390741000000,0.0187169400000000,-
0.0465116280000000;
    -0.0126682260000000,0.0252654710000000,-0.0461986990000000,-
0.0237419350000000,0.154159869000000,0.0226537220000000,-
0.0237419350000000,-0.0516717330000000,0.0171748100000000,-
0.0731707320000000;
    -
0.00121538500000000,0.0446428570000000,0.00681871600000000,0.00211472400000
000,-0.0155477030000000,0.107594937000000,0.00211472400000000,-
0.0160256410000000,-0.0210732980000000,-0.0263157890000000;
    -0.00787110500000000,0.0769230770000000,-0.0366651100000000,-
0.0308625690000000,-0.113424264000000,-0.00571428600000000,-
0.0308625690000000,-0.0358306190000000,-0.0299505280000000,-
0.0540540540000000;

0.00523210700000000,0.178571429000000,0.0729696970000000,0.0204137180000000
,0.0700404860000000,0.0890804600000000,0.0204137180000000,488.864864900000,
0.00110268800000000,0;];
P=P'; %%P: N-by-T
r=P';
Qi=riskI_envelope( I,N,T );
rj=r(:,J);
cova=mean(rj.*Qi);

end
```

5.// CVaR_asset_I.m

```
function [ CVaR_I ] = CVaR_asset_I( I,N,T )
%UNTITLED Summary of this function goes here
%   Detailed explanation goes here
risk_envlopI=riskI_envelope( I,N,T );
%%%
givePortfoval=portfolio_given(N,T);   % an given vector over T periods
%%%
CVaR_I=(givePortfoval*(1-risk_envlopI))/T;

end
```

6.// CVaR_optimality_condition.m

```
function [ XV ] = CVaR_optimality_condition( N,T )
%OPTIMALITY_CONDITION Summary of this function goes here
%   Detailed explanation goes here
c=Mix_CVaR_coeff( N,T);  %% the matrix used for calculate independent
variables xi
C=c;
d=zeros(T-1,1);
Aeq=ones(1,T-1);
beq=1;
lb=zeros(T-1,1);
ub=ones(T-1,1);
%%using 'linsolve' _Solve linear system of equations given in matrix form
%%[X,R] = linsolve(A,B);
X = lsqlin(C,d,[],[],Aeq,beq,lb,ub);

K_index=find(X>=0);
XV=[];
for i=1:numel(K_index)
     t=K_index(i);
     XV=[XV,X(t)];   %% or use command XV(i)=X(t);;;
end

%%%another method using the old definition of 'linsolve' by constructing
%%%symbolic matrices;;;;however, in the new release version, no definition
%%%of solver 'linsolve' for input arguments of type 'sym';;
%%%Therefore, we use 'solve' iteratively ,see another m_file;;;;
end
```

7.// mean_returnOfobservation.m

```
function [  mean_return ] = mean_returnOfobservation( N,T)
%MEAN_RETURNOFOBSERVATION Summary of this function goes here
%%N=10,T=11,P:(T-by-N)
%   mean_return vector obtained from observation over T time periods
P = [0.0162355710000000,-0.00246812000000000,-0.0404016810000000,-
0.0419082760000000,-0.0737995490000000,-0.0340909090000000,-
0.0419082760000000,1645.88427300000,0.0354098360000000,-0.184210526000000;
```

```
    0.0257420960000000,-
0.00515463900000000,0.0194694570000000,0.0707015130000000,-
0.00939457200000000,0.0745098040000000,0.0707015130000000,0.171171171000000
,0.0232742240000000,-0.0645161290000000;
    0.0265704650000000,0.0130569950000000,0.00381952700000000,-
0.00539568300000000,-0.0386371620000000,-0.0364963500000000,-
0.00539568300000000,0.107692308000000,0.0479653410000000,0.0689655170000000;
    0.0131094440000000,-
0.0180032730000000,0.0623067780000000,0.0211831570000000,0.0372670810000000
,0.121212121000000,0.0211831570000000,0.00694444400000000,0.022442049000000
0,0.322580645000000;
    -
0.0114467550000000,0.0729166670000000,0.0315648090000000,0.0131545660000000
,-0.126805213000000,0.0405405410000000,0.0131545660000000,-
0.999508966000000,-0.0127075810000000,0.121951220000000;
    0.0142033320000000,0.106796117000000,-0.0108506940000000,-
0.00549313400000000,-0.0338846310000000,0.0324675320000000,-
0.00549313400000000,1515.85393300000,0.0783969580000000,-0.0652173910000000;
-0.00970231200000000,-0.0417543860000000,-0.0217200530000000,-
0.0271152400000000,0.0237995820000000,-0.0283018870000000,-
0.0271152400000000,-0.999390741000000,0.0187169400000000,-
0.0465116280000000;
    -0.0126682260000000,0.0252654710000000,-0.0461986990000000,-
0.0237419350000000,0.154159869000000,0.0226537220000000,-
0.0237419350000000,-0.0516717330000000,0.0171748100000000,-
0.0731707320000000;
    -
0.00121538500000000,0.0446428570000000,0.00681871600000000,0.00211472400000
000,-0.0155477030000000,0.107594937000000,0.00211472400000000,-
0.0160256410000000,-0.0210732980000000,-0.0263157890000000;
    -0.00787110500000000,0.0769230770000000,-0.0366651100000000,-
0.0308625690000000,-0.113424264000000,-0.00571428600000000,-
0.0308625690000000,-0.0358306190000000,-0.0299505280000000,-
0.0540540540000000;

0.00523210700000000,0.178571429000000,0.0729696970000000,0.0204137180000000
,0.0700404860000000,0.0890804600000000,0.0204137180000000,488.864864900000,
0.00110268800000000,0;];
r=P;
ri1=r(:,1);
mean_return=mean(ri1);   % average of observations over T periods for asset
1;;
for i=2:N
   ri=r(:,i);
   mean_return=[mean_return,mean(ri)];
end
end
```

8.// Mix_CVaR_coeff.m

```
function [ c ] = Mix_CVaR_coeff( N,T )
%MIX_CVAR_COEFF Summary of this function goes here
%   Detailed explanation goes here
%%%
mean_return=mean_returnOfobservation( N,T);

%%minimum desired return;; % risk_free fixed saving account interest rate
%%choose fixed saving rate 2.03% for one year, we need weekly rate;;
rfm=0.0203;    % interest rate for one year
```

```
rfm=(rfm+1)^(7/365)-1 ; % weekly risk_free interst rate chosen
%%%
r0=0;
%CVaR_I=CVaR_asset_I( I,N,T );
%%%
%%%
givePortfoval=portfolio_given(N,T);   % an given vector over T periods
mean_givenPortfoval=mean(givePortfoval);
%%%
%cova=Covar_asset_I( I,N,T )
%%%
for j=1:N
   for i=1:T-1
       cova=Covar_asset_I(i,j,N,T );
       CVaR_I=CVaR_asset_I( i,N,T );
       c(i,j)=(mean_return(j)-r0)*CVaR_I+(mean_givenPortfoval-r0)*cova;
   end
end
end
```

9.// riskI_envelope.m

```
function [ xI ] = riskI_envelope( I,N,T )
%RISK_I_ENVELOPE Summary of this function goes here
%%%
mean_return=mean_returnOfobservation(N,I);
%%%
givePortfoval=portfolio_given(N,T);   % an given vector over T periods
mean_givenPortfoval=mean(givePortfoval);
%%%
nAssets=numel(mean_return);

%%%parameters
apha=[];
for i=1:T-1        %% here, T-1=N ,denotes the number of assets
       apha=[apha,i/T];
end
aphI=apha(I);
%%%
%%% optimization using linear programming solver
f=givePortfoval;
Aeq=ones(1,T);
beq=1;
lb=zeros(T,1);
ub=(1/aphI)*ones(T,1);
[xI, fval]=linprog(f,[],[],Aeq,beq,lb,ub);
End
```

10.// risk_profile.m

```
function [ G,I_lamda ] = risk_profile( N,T )
%RISK_PROFILE Summary of this function goes here
%   Detailed explanation goes here
%%desired return;; % risk_free fixed saving account interest rate
%%choose fixed saving rate 2.03% for one year, we need weekly rate;;
rfm=0.0203;    % interest rate for one year
rfm=(rfm+1)^(7/365)-1 ; % weekly risk_free interst rate chosen
%%%
lamda=CVaR_optimality_condition( N,T );
```

```
num_lamda=numel(lamda);
h=1/(num_lamda-1);
alpha=0:h:1;
%%%
%givePortfoval=portfolio_given(N,T);    % an given vector over T periods
%%%
      g(1)=0;
      us=lamda./alpha;
  for i=2:num_lamda
      s1=sum(lamda(1:i));       %%%!!!
      ai=alpha(i);
      s2=ai*sum(us(i:num_lamda));       %%%
      g(i)=s1+s2-ai;
  end
      g(num_lamda)=0;
  xi = 0:.025:1;
  yi = interp1(alpha,g,xi);
  plot(alpha,g,'o',xi,yi);
end
```

11.// CVaR_optimality_condition_2.m

```
function [ output_args ] = CVaR_optimality_condition_2( N,T )
%UNTITLED Summary of this function goes here
%   Detailed explanation goes here
%%%another method using the old definition of 'linsolve' by constructing
%%%symbolic matrices;;;;however, in the new release version, no definition
%%%of solver 'linsolve' for input arguments of type 'sym';;
%%%Therefore, we use 'solve' iteratively
%%%to deal with input arguments of 'sym' type;;;
%%%generating a symbolic vector with n variables
xi=[];
for i=1:T-1
    t=sym(['x' int2str(i)],'positive');
    xi=[xi;t];
end
S=[];
for i=1:N
       t=sym(sum(c(:,i).*xi));
       S=[S;t];
end
%%additional parameters,i.e. additional requirement for solution
adt=sym(sum((Aeq').*xi));
beq=zeros(N,1);
B=[beq;ones];

beq=1
Bt=[];
for i=1:T-1
    Bt=[Bt;zeros(1,T)]
end
adB=zeros(1,T)
adB(T)=beq;
B=[Bt;adB]
end
```

Appendix IV

1.// deviation_measure_unique.m

```
function [ x,fval ] = deviation_measure_unique( I,N,T )
%UNTITLED Summary of this function goes here
%   evaluate at i/T;;;
%%desired return;; % risk_free fixed saving account interest rate
%%choose fixed saving rate 2.03% for one year, we need weekly rate;;
rfm=0.0203;    % interest rate for one year
rfm=(rfm+1)^(7/365)-1 ; % weekly risk_free interst rate chosen
%%%
%%%% average of observations over T periods for N asset;
mean_return=mean_returnOfobservation( N,T);
%%%
%%%
%%mean_return=[rfm,mean_return];
CVaR_I=CVaR_asset_I( I,N,T );   %%the number of given portfolio weights is
N;;
%%%
lamd_coeff=mean_return;
%%%
f=[zeros(1,N),0,zeros(1,T),1];
A1=[lamd_coeff,-1,(1/I)*ones(1,T),-CVaR_I];
%%%
P = [0.0162355710000000,-0.00246812000000000,-0.0404016810000000,-
0.0419082760000000,-0.0737995490000000,-0.0340909090000000,-
0.0419082760000000,1645.88427300000,0.0354098360000000,-0.184210526000000;
    0.0257420960000000,-
0.00515463900000000,0.0194694570000000,0.0707015130000000,-
0.00939457200000000,0.0745098040000000,0.0707015130000000,0.171171171000000
,0.0232742240000000,-0.0645161290000000;
    0.0265704650000000,0.0130569950000000,0.00381952700000000,-
0.00539568300000000,-0.0386371620000000,-0.0364963500000000,-
0.00539568300000000,0.107692308000000,0.0479653410000000,0.0689655170000000;
    0.0131094440000000,-
0.0180032730000000,0.0623067780000000,0.0211831570000000,0.0372670810000000
,0.121212121000000,0.0211831570000000,0.00694444400000000,0.022442049000000
0,0.322580645000000;
    -
0.0114467550000000,0.0729166670000000,0.0315648090000000,0.0131545660000000
,-0.126805213000000,0.0405405410000000,0.0131545660000000,-
0.999508966000000,-0.0127075810000000,0.121951220000000;
    0.0142033320000000,0.106796117000000,-0.0108506940000000,-
0.00549313400000000,-0.0338846310000000,0.0324675320000000,-
0.00549313400000000,1515.85393300000,0.0783969580000000,-0.0652173910000000;
-0.00970231200000000,-0.0417543860000000,-0.0217200530000000,-
0.0271152400000000,0.0237995820000000,-0.0283018870000000,-
0.0271152400000000,-0.999390741000000,0.0187169400000000,-
0.0465116280000000;
    -0.0126682260000000,0.0252654710000000,-0.0461986990000000,-
0.0237419350000000,0.154159869000000,0.0226537220000000,-
0.0237419350000000,-0.0516717330000000,0.0171748100000000,-
0.0731707320000000;
    -
0.00121538500000000,0.0446428570000000,0.00681871600000000,0.00211472400000
000,-0.0155477030000000,0.107594937000000,0.00211472400000000,-
0.0160256410000000,-0.0210732980000000,-0.0263157890000000;
    -0.00787110500000000,0.0769230770000000,-0.0366651100000000,-
0.0308625690000000,-0.113424264000000,-0.00571428600000000,-
0.0308625690000000,-0.0358306190000000,-0.0299505280000000,-
0.0540540540000000;
```

```
0.00523210700000000,0.178571429000000,0.0729696970000000,0.0204137180000000
,0.0700404860000000,0.0890804600000000,0.0204137180000000,488.864864900000,
0.00110268800000000,0;];
P=P';
r=P';
%%%
Z_coef=(-1)*eye(T);
A2=[r,ones(T,1),Z_coef,zeros(T,1)];
%%%
A3=[zeros(T,N),zeros(T,1),Z_coef,zeros(T,1)];
A4=[-lamd_coeff,0,zeros(1,T),0];
A5=[(-1)*eye(N),zeros(N,1),zeros(N,T),zeros(N,1)];
A=vertcat(A1,A2,A3,A4,A5);
b=vertcat(0,zeros(T,1),zeros(T,1),zeros(N,1),-rfm);
Aeq=[ones(1,N),0,zeros(1,T),0];
beq=ones;
%lb=zeros(N+3,1);
%ub=ones(N+3,1);
%lb(N+2:N+3)=[];   %%%%
%ub(N+1:N+3)=[];
[x,fval] = linprog(f,A,b,Aeq,beq);

end
```

Asset chosen_I: N=3, T=4

	FTSE 100 (^FTSE)-Historical Prices(Weekly)-[from 08/07/2013 to 05/08/2013]							**Rate of Return**
	Date	**Open**	**High**	**Low**	**close**	**Avg Vol**	**Adj Close***	
	08-Jul-13	6,375.50	6,585.70	6,375.50	6,544.90	626,287,700	6,544.90	0.013109444
	15-Jul-13	6,544.90	6,657.40	6,517.40	6,630.70	690,582,700	6,630.70	-0.011446755
	22-Jul-13	6,630.70	6,662.20	6,535.20	6,554.80	546,564,700	6,554.80	0.014203332
	29-Jul-	6,554.80	6,696.60	6,544.10	6,647.90	714,583,200	6,647.90	-0.0065735

	13							04
	05-Aug-13	6,647.90	6,684.90	6,562.30	6,604.20	652,477,400	6,604.20	
	SMI (^SSMI)-VTX -Historical Prices(Weekly)-[from 08/07/2013 to 05/08/2013]							**Rate of Return**
	Date	**Open**	**High**	**Low**	**close**	**Avg Vol**	**Adj Close***	
	08-Jul-13	7,857.90	8,041.30	7,843.30	7,983.20	58,763,400	7,983.20	-0.006901994
	15-Jul-13	7,999.50	8,031.50	7,853.50	7,928.10	61,699,100	7,928.10	-0.016561345
	22-Jul-13	7,950.00	7,979.50	7,777.00	7,796.80	72,467,200	7,796.80	0.021431869
	29-Jul-13	7,803.70	7,974.40	7,760.50	7,963.90	37,350,300	7,963.90	0.004131142
	05-Aug-13	7,973.50	8,080.30	7,963.10	7,996.80	33,939,600	7,996.80	
	IBEX 35 (^IBEX)-MCE-Historical Prices(Weekly)-[from 08/07/2013 to 05/08/2013]							**Rate of Return**
	Date	**Open**	**High**	**Low**	**close**	**Avg Vol**	**Adj Close***	
	08-Jul-	7,922.00	8,111.50	7,800.80	7,844.70	196,434,900	7,844.70	0.012556248

	13							
	15-Jul-13	7,832.40	7,978.90	7,702.70	7,943.20	174,211,200	7,943.20	0.051666835
	22-Jul-13	7,954.90	8,395.90	7,917.10	8,353.60	211,778,100	8,353.60	0.026383835
	29-Jul-13	8,374.60	8,578.30	8,349.10	8,574.00	179,804,700	8,574.00	-0.00519011
	05-Aug-13	8,585.10	8,617.30	8,469.90	8,529.50	139,952,800	8,529.50	
The constructed portfolio of rate of return in T=4 scenarios and N=3 assets								
		T=1	T=2	T=3	T=4			
	N=1	0.01310944	-0.01144675	0.01420333	-0.006573504			
	N=2	-0.00690199	-0.01656134	0.02143186	0.004131142			
	N=3	0.01255624	0.05166683	0.02638383	-0.00519011			
The given Portfolio the investor holds								

(lamda)								
	N=1	N=2	N=3					
Assets weight	0.24	0.35	0.41					

Asset chosen_II: N=10, T=11

	Historical Prices(Weekly)-[from 17/06/2013 to 02/09/2013]								
	1. FTSE 100 (^FTSE)-FTSE						Rate of Return		
Date	Open	High	Low	Close	Volume	Adj Close			
17/06/2013	6308.3	6397.3	6116.2	6116.2	867291700	6116.2	0.016235571		
24/06/2013	6116.2	6271.8	6023.4	6215.5	826128600	6215.5	0.025742096		
01/07/2013	6215.5	6498.6	6185.2	6375.5	686155200	6375.5	0.026570465		
08/07/2013	6375.5	6585.7	6375.5	6544.9	626287700	6544.9	0.013109444		
15/07/2013	6544.9	6657.4	6517.4	6630.7	690582700	6630.7	-0.011446755		
22/07/2013	6630.7	6662.2	6535.2	6554.8	546564700	6554.8	0.014203332		
29/07/2013	6554.8	6696.6	6544.1	6647.9	714583200	6647.9	-0.009702312		

05/08/2013	6647.9	6684.9	6507.2	6583.4	629077200	6583.4	-0.012668226
12/08/2013	6583.4	6628	6460.1	6500	609398000	6500	-0.001215385
19/08/2013	6500	6516.7	6386.7	6492.1	536755100	6492.1	-0.007871105
27/08/2013	6492.1	6494.2	6424	6441	760395800	6441	0.005232107
02/09/2013	6412.9	6532.3	6412.9	6474.7	683841700	6474.7	#REF!
	2. GW Pharmaceuticals PLC (GWP.L)-LSE						Rate of Return
Date	Open	High	Low	Close	Volume	Adj Close	
17/06/2013	45.5	50	45	48.62	77000	48.62	-0.00246812
24/06/2013	47	49.7	46.75	48.5	67900	48.5	-0.005154639
01/07/2013	48.98	49.9	46.5	48.25	44800	48.25	0.013056995
08/07/2013	48	50	47	48.88	372000	48.88	-0.018003273
15/07/2013	49.17	49.5	48	48	35600	48	0.072916667
22/07/2013	49.75	51.65	48	51.5	56900	51.5	0.106796117
29/07/2013	51	58	50.91	57	275000	57	-0.041754386
05/08/2013	55	59.67	53.05	54.62	86600	54.62	0.025265471

12/08/2013	53	58	53	56	225500	56	0.044642857
19/08/2013	55.75	59	54	58.5	80500	58.5	0.076923077
26/08/2013	58.5	63	54.75	63	118900	63	0.178571429
02/09/2013	62	74.5	60.55	74.25	230800	74.25	#REF!
	3.DIXONS RETAIL PLC ORD 2.5P (DXNS.L)-LSE						Rate of Return
Date	Open	High	Low	Close	Volume	Adj Close	
17/06/2013	42	44.75	41.55	42.82	18432100	42.82	-0.040401681
24/06/2013	42.5	42.66	36.55	41.09	22460700	41.09	0.019469457
01/07/2013	41.1	42.3	40.75	41.89	14492600	41.89	0.003819527
08/07/2013	41.9	44.12	40.67	42.05	16475300	42.05	0.062306778
15/07/2013	42.06	45.13	42	44.67	11078500	44.67	0.031564809
22/07/2013	44.68	46.78	44.5	46.08	12980700	46.08	-0.010850694
29/07/2013	46.09	46.99	44.8	45.58	8765900	45.58	-0.021720053
05/08/2013	45.59	46.61	44.1	44.59	8979200	44.59	-0.046198699
12/08/2	44.6	44.82	40.7	42.53	11239700	42.53	0.006818716

013							
19/08/2013	42.52	43.74	40.5	42.82	9890800	42.82	-0.03666511
26/08/2013	42.82	43	40.52	41.25	9212200	41.25	0.072969697
02/09/2013	41.38	44.97	41.35	44.26	18619400	44.26	#REF!
	4. ICAP PLC (IAP.L)-LSE						Rate of Return
Date	Open	High	Low	Close	Volume	Adj Close	
17/06/2013	358.5	384.4	358.5	379.4	2324400	364.29	-0.041908276
24/06/2013	379.3	386.8	343	363.5	2847100	363.5	0.070701513
01/07/2013	365.4	407.4	362.5	389.2	1535900	389.2	-0.005395683
08/07/2013	391.2	406.5	366.6	387.1	2173000	387.1	0.021183157
15/07/2013	388.5	395.9	381.2	395.3	1075100	395.3	0.013154566
22/07/2013	395.3	409.8	395.3	400.5	855200	400.5	-0.005493134
29/07/2013	401.2	412.4	394.4	398.3	1297100	398.3	-0.02711524
05/08/2013	398.3	406.3	385.09	387.5	871500	387.5	-0.023741935
12/08/2013	388.6	389.4	374.5	378.3	874700	378.3	0.002114724

19/08/2013	378.2	381.1	366.6	379.1	795600	379.1	-0.030862569
26/08/2013	379.1	379.1	364.1	367.4	712300	367.4	0.020413718
02/09/2013	369.3	380.3	365	374.9	775700	374.9	#REF!
	5. Imagination Technologies Group PLC (IMG.L)-LSE						Rate of Return
Date	Open	High	Low	Close	Volume	Adj Close	
17/06/2013	307.8	329.7	281.98	310.3	2265300	310.3	-0.073799549
24/06/2013	310	311.9	271.1	287.4	1821800	287.4	-0.009394572
01/07/2013	289.4	302.4	274.6	284.7	1086800	284.7	-0.038637162
08/07/2013	287.7	295.7	271.7	273.7	1107700	273.7	0.037267081
15/07/2013	277.3	291.67	271.8	283.9	944500	283.9	-0.126805213
22/07/2013	285.1	289.98	244.39	247.9	2100800	247.9	-0.033884631
29/07/2013	249.3	251.7	234.5	239.5	1172200	239.5	0.023799582
05/08/2013	239.3	249.5	235	245.2	578600	245.2	0.154159869
12/08/2013	246	287.3	242.8	283	1242000	283	-0.015547703
19/08/2	283.1	285.2	260.5	278.6	692800	278.6	-0.113424264

013							
26/08/2013	278.6	278.6	241.18	247	626100	247	0.070040486
02/09/2013	250.7	268.07	247.9	264.3	690000	264.3	#REF!
	6. Wincanton PLC (WIN.L)-LSE						Rate of Return
Date	Open	High	Low	Close	Volume	Adj Close	
17/06/2013	67.5	68.75	64.75	66	187400	66	-0.034090909
24/06/2013	66	67.75	63.2	63.75	338200	63.75	0.074509804
01/07/2013	65	68.5	62.67	68.5	243600	68.5	-0.03649635
08/07/2013	67.5	69.5	66	66	78200	66	0.121212121
15/07/2013	66	75	65	74	236700	74	0.040540541
22/07/2013	74.5	78.7	72	77	268400	77	0.032467532
29/07/2013	74.25	81	73.12	79.5	231800	79.5	-0.028301887
05/08/2013	80	80.25	73.25	77.25	161600	77.25	0.022653722
12/08/2013	78	81.75	77.25	79	143100	79	0.107594937
19/08/2013	79	89	79	87.5	267700	87.5	-0.005714286

26/08/2013	87.5	88.86	84.65	87	94200	87	0.08908046
02/09/2013	88	94.9	86.5	94.75	155800	94.75	#REF!
	7. ICAP PLC (IAP.L)-LSE						Rate of Return
Date	Open	High	Low	Close	Volume	Adj Close	
17/06/2013	358.5	384.4	358.5	379.4	2324400	364.29	-0.041908276
24/06/2013	379.3	386.8	343	363.5	2847100	363.5	0.070701513
01/07/2013	365.4	407.4	362.5	389.2	1535900	389.2	-0.005395683
08/07/2013	391.2	406.5	366.6	387.1	2173000	387.1	0.021183157
15/07/2013	388.5	395.9	381.2	395.3	1075100	395.3	0.013154566
22/07/2013	395.3	409.8	395.3	400.5	855200	400.5	-0.005493134
29/07/2013	401.2	412.4	394.4	398.3	1297100	398.3	-0.02711524
05/08/2013	398.3	406.3	385.09	387.5	871500	387.5	-0.023741935
12/08/2013	388.6	389.4	374.5	378.3	874700	378.3	0.002114724
19/08/2013	378.2	381.1	366.6	379.1	795600	379.1	-0.030862569
26/08/2013	379.1	379.1	364.1	367.4	712300	367.4	0.020413718

02/09/2013	369.3	380.3	365	374.9	775700	374.9	#REF!
	8. JPMorgan Indian Investment Trust PLC (JII.L)-LSE						Rate of Return
Date	Open	High	Low	Close	Volume	Adj Close	
17/06/2013	3.54	4.08	3.35	3.37	302600	3.37	1645.884273
24/06/2013	4000	6500	3400	5550	11000	5550	0.171171171
01/07/2013	3.57	7000	3.45	6500	85800	6500	0.107692308
08/07/2013	5900	7500	3.63	7200	28000	7200	0.006944444
15/07/2013	3.69	8250	3.68	7250	47400	7250	-0.999508966
22/07/2013	6720	6775	3.56	3.56	16700	3.56	1515.853933
29/07/2013	3.55	5837.5	3.37	5400	64700	5400	-0.999390741
05/08/2013	5000	5700	3.23	3.29	164600	3.29	-0.051671733
12/08/2013	3.29	3.37	3.11	3.12	243200	3.12	-0.016025641
19/08/2013	3.1	3.11	2.86	3.07	485100	3.07	-0.035830619
27/08/2013	2.99	2.99	2.67	2.96	388900	2.96	488.8648649
02/09/2	2.99	1101	2.84	1450	194100	1450	#REF!

013							
	9. Lloyds Banking Group PLC (LLOY.L)-LSE						Rate of Return
Date	Open	High	Low	Close	Volume	Adj Close	
17/06/2013	61.48	62.97	60.57	61	152569800	61	0.035409836
24/06/2013	60.84	64.28	59.76	63.16	163240200	63.16	0.023274224
01/07/2013	63.85	66.52	61.68	64.63	143177000	64.63	0.047965341
08/07/2013	65.6	68.46	65.3	67.73	150164900	67.73	0.022442049
15/07/2013	68	70.51	67.97	69.25	149364900	69.25	-0.012707581
22/07/2013	69.28	70.04	67.78	68.37	94385300	68.37	0.078396958
29/07/2013	68.51	75.06	66.51	73.73	196988100	73.73	0.01871694
05/08/2013	74.5	77.36	73.35	75.11	159935200	75.11	0.01717481
12/08/2013	75.17	77.39	70.25	76.4	137358100	76.4	-0.021073298
19/08/2013	76.36	76.76	72.76	74.79	115211200	74.79	-0.029950528
26/08/2013	74.79	74.79	71.58	72.55	94957900	72.55	0.001102688
02/09/2013	73.12	74.04	71.98	72.63	95907600	72.63	#REF!
	10. PATERNOSTER RESOURCES PLC ORD 0						Rate of Return

	(PRS.L)-LSE						
Date	Open	High	Low	Close	Volume	Adj Close	
17/06/2013	0.56	0.56	0.31	0.38	19474800	0.38	-0.184210526
24/06/2013	0.39	0.4	0.3	0.31	4251600	0.31	-0.064516129
01/07/2013	0.33	0.34	0.27	0.29	7042300	0.29	0.068965517
08/07/2013	0.3	0.3	0.25	0.31	3832600	0.31	0.322580645
15/07/2013	0.3	0.44	0.27	0.41	9453200	0.41	0.12195122
22/07/2013	0.38	0.5	0.35	0.46	7516600	0.46	-0.065217391
29/07/2013	0.43	0.5	0.37	0.43	6064900	0.43	-0.046511628
05/08/2013	0.4	0.64	0.37	0.41	28419400	0.41	-0.073170732
12/08/2013	0.38	0.41	0.33	0.38	13738800	0.38	-0.026315789
19/08/2013	0.36	0.39	0.31	0.37	3849600	0.37	-0.054054054
26/08/2013	0.37	0.38	0.31	0.35	3506500	0.35	0
02/09/2013	0.36	0.36	0.33	0.35	314600	0.35	#REF!
	11.Halfords Group PLC (HFD.L)-LSE						Rate of Return

Date	Open	High	Low	Close	Volume	Adj Close			
17/06/2013	321.7	322	302	302.4	717800	293.72	0.049272487		
24/06/2013	301.5	319.6	296.5	317.3	633900	308.19	-0.015442799		
01/07/2013	318.8	325.76	303	312.4	637200	312.4	0.139884763		
08/07/2013	315	363.23	313.2	356.1	1324400	356.1	0.049143499		
15/07/2013	356.2	380.4	353.7	373.6	699400	373.6	-0.012312634		
22/07/2013	373.4	374.6	362.7	369	453400	369	0.005149051		
29/07/2013	374.4	375.1	363.8	370.9	423100	370.9	-0.01698571		
05/08/2013	373.2	381.8	363.3	364.6	413600	364.6	-0.020844761		
12/08/2013	366.1	372.2	350.04	357	453400	357	0.072268908		
19/08/2013	359.2	388.5	357.5	382.8	412100	382.8	-0.054336468		
26/08/2013	382.8	388.6	361.05	362	455400	362	0.011878453		
02/09/2013	362.8	369.2	358.44	366.3	369600	366.3	#REF!		